SAINT MICHAEL COLLEGE OF CARAGA

AGRILINK: AN E-COMMERCE PLATFORM FOR AGRICULTURAL MARKET LINKAGE IN BUENAVISTA, AGUSAN DEL NORTE

A Thesis Presented To
The Faculty of
College of Computing and Information Sciences

In Partial Requirements for the Degree
BACHELOR OF SCIENCE IN INFORMATION AND TECHNOLOGY

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March 2026

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ACKNOWLEDGEMENT

Researchers sincerely expressed their gratitude to all who contributed to the success of this Capstone project, "AgriLink: An E-commerce Platform for Agricultural Market Linkage in Buenavista, Agusan del Norte"

The Researchers thank Mr. Jessie S. Mahinay, our counsellor, for his guidance, support, and valuable insights into the entire study. Our appreciation also extends to the Department of Agriculture in Buenavista, Agusan del Norte and all the staff.

Special thanks to our panel members, family, and friends for their encouragement, understanding, and unwavering support, which have inspired us to complete this work.

Above all, we are grateful to the Almighty God for His wisdom and blessings that have made this project possible.

With sincere gratitude,

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CHAPTER 1

INTRODUCTION

1.1 Project Context

E-commerce websites are utilized in nearly every aspect of online business. Even in farming, farmers' e-commerce websites are gaining popularity, enabling them to sell directly to customers without intermediaries. Not only does this benefit farmers by increasing their revenues, but it also allows consumers to access fresh, locally produced products. However, it does not appear that all agricultural offices are adopting this type of digitalization in their operations, which could slow down the growth and efficiency of the farming industry. With the adoption of e-commerce platforms, farmers can expand their markets and streamline their sale processes, ultimately serving both producers and consumers.

As [1] posited, farm marketplaces remain basic, disorganized, and unplanned technological interventions; hence, the buyers and sellers are caught in prolonged, manual procedures. The lack of synergistic e-commerce platforms creates insurmountable barriers for smallholder farmers, making it impossible for them to obtain fair prices, timely market data, and quick access to consumers, and eliminating the need for intermediaries. In the same way, [9] described that the use of e-commerce among rural farm communities has great potential to enhance farmers' ability to sell. Online marketplaces enable farmers to access larger markets beyond their communities, reduce their reliance on mediators who take large margins, and communicate directly with consumers. It not only streamlines trading but also enables farmers to receive improved prices, enhances their incomes, and offers them greater opportunities to diversify. In the long term, e-commerce has enabled farmers to become more competitive, sustainable, and resilient in the emerging agricultural market.

Further, [2] points out that through their inclusion in digitalization, farmers are enabled to have the financial capacity for sustainable agriculture through the integration of financial services such as microloans and mobile banking. Such financing will be instrumental in facilitating farmers'

access to modern technology and sustainable agricultural practices, enabling them to become more productive and profitable. The merging of digitalization with agriculture can turn the industry towards greater sustainability.

Today, most Buenavista Agusan del Norte sales take place in open markets or trading centers. Despite it being the old way for decades, this is slow, costly, and risky. Farmers can drive for hours to the market, only to return without having sold a single item. Additionally, there is no direct way to establish beforehand how much consumers are willing to pay for their goods or what price they are willing to pay.

To manage these challenges, the proposed solution is AgriLink, an electronic marketplace that enables farmers to deal directly with buyers. Farmers would be able to sell their produce online, and the buyers could place orders, pay electronically, and organize delivery more efficiently. As [4] suggests, electronic platforms and electronic commerce systems make things more efficient by providing easier access to markets, reducing transaction costs, and offering farmers better price information. These systems can help small farmers receive better prices at the farm gate, fair trade, and a more sustainable local economy.

Through AgriLink, Buenavista's farming community will have access to an open, fair, and transparent market. The system eliminates waste elimination, reduces reliance on intermediators, and ensures that farmers receive a fair return for their efforts, while customers enjoy fresher produce and more convenient transactions.

1.2 Review of Related Literature

This chapter presents the related literature and studies relevant to the researchers' project on e-commerce for agricultural market linkage. It emphasizes the need to implement the proposed platform, its potential benefits to farmers and buyers, and the issues it seeks to address. The related literature is gathered from electronic materials, e-books, websites, scholarly articles, and peer-reviewed journals.

E-Commerce as an Agricultural Market Linkage

E-commerce has also proven to be an effective platform for connecting farmers directly with consumers, removing intermediaries, and increasing their income sources. In the Philippines, [1] showed how internet marketing in Quirino Province improved market access and earnings for smallholder farmers. Similarly, [3] highlighted that e-platforms in Sub-Saharan Africa foster more fair pricing, support sustainable farming practices, and boost market participation. Overall, these studies suggest that e-commerce has the potential to transform traditional agri-food systems into more inclusive and transparent markets.

Barriers and Challenges in Implementing Agricultural E-Commerce

Despite its promise, the full adoption of e-commerce in agriculture is hindered by several challenges. [4] showed that Philippine smallholder farmers are constrained by digital literacy, and there is insufficient institutional support for online business. [4] It is also noted that, despite the growth of digital infrastructures, hesitation among farmers persists due to concerns about trust, inadequate training, and low awareness. These findings suggest that technical support should be combined with behavioral and learning interventions to achieve the successful adoption of e-commerce.

Digital Transformation and Inclusiveness

Digital technologies have been demonstrated to improve the inclusiveness of agriculture by connecting smallholders with domestic and international markets. [5] indicated that farmers in the

Philippines became more receptive to digital platforms during the COVID-19 pandemic, viewing them as essential to maintaining a market presence. [6] cited that e-commerce usage improves productivity and competitiveness in the Philippine agriculture sector. Similarly, [7] illustrated that e-commerce is broadening participation in the Malaysian agri-food market by enhancing access and competitiveness. Digital inclusiveness is often unbalanced, particularly among resource-poor farmers who lack adequate exposure to technology.

Infrastructure and System Development

Infrastructure development is necessary to sustain agricultural e-commerce. [8] emphasized that well-functioning logistics and rural digital infrastructure are critical to enabling farmers' participation in online markets. [9] It was also found that digital backbone systems are crucial for integrating smallholders into larger supply chains. Samer and Raman [10] supported this assumption by claiming that successful e-farming platforms must have reliable internet connectivity, technology skill training, and efficient supply chain coordination. Likewise, [[11] demonstrated that Thailand's agricultural e-commerce platforms were effective when combined with enabling policy frameworks and online infrastructure. Such papers confirm that infrastructure, technology, and policy integration are the keystones of an effective digital farming system.

Emerging Technologies and System Innovations in Agricultural E-Commerce

New technological breakthroughs have enabled the development of more intelligent, more efficient digital trading networks. [12] introduced an IoT-driven model of e-commerce that incorporates cloud-based data systems to enhance coordination between producers and markets. Abao et al. [13] developed a peer-to-peer (P2P) agricultural e-commerce platform allowing for direct buy-and-sell transactions between farmers and consumers. Overall, the research highlights the increasing role of technology-driven innovations in improving agricultural e-commerce, making it more transparent, efficient, and responsive to user needs.

1.3 Project Objectives

The researchers' study aims to design and develop an e-commerce platform for agricultural market linkage under the Department of Agriculture – Buenavista, Agusan del Norte, named AgriLink.

Specifically, it aims to:

1. Integrate a report generation feature within the system that automatically converts dashboard data into downloadable reports in Excel format for documentation and analysis purposes.
2. Develop key system features such as product listing, order management, and delivery tracking to streamline agricultural transactions.
3. Implement secure transaction mechanisms by enabling direct farmer-to-buyer exchanges through cash-on-delivery and pickup-based processes, and by integrating security features as password hashing to protect user accounts and data.
4. Evaluate the system using ISO 25010 Standards to ensure its functionality, efficiency, and usability.

1.4 Scope and Limitation

The theme of this study is the design and development of AgriLink, which is an e-commerce platform for Agricultural Market Linkage in Buenavista, Agusan del Norte. The system is designed to connect farmers, buyers, and traders in the area through a centralized e-commerce platform rather than traditional manual linkages. Its operational breadth encompasses user registration and authentication, product listing and administration, order placement, digital payment integration, delivery tracking, messaging, and administrative reporting. Authorized users will be provided with a secure account, while the administrator oversees registration, administers product listings, monitors transactions, and generates reports. The platform is primarily designed to serve direct consumers rather than large-scale wholesalers or distributors. To support this focus and manage logistics for

individual consumer, the system will include a feature to limit the number of kilos that can be purchased per order. The coverage is limited to Buenavista stakeholders, allowing the system to provide an immediate solution to the issues and problems encountered by farmers within the area, in line with the primary objective of this study.

Notwithstanding these features, the platform faces various limitations. It requires an active internet connection to operate and can be accessed on compatible devices, such as smartphones, tablets, desktops, or laptops, using supported applications or browsers. Logistics and physical delivery are not part of the system; these remain at the user's own discretion. Payment processing is limited, however, to integrated channels like e-wallets and bank transfers. Additionally, the research will be conducted during the pilot phase, employing a limited number of participants from Buenavista, thereby limiting its representativeness. These constraints relate to the second and fourth objectives, which involve establishing system characteristics and evaluating performance in accordance ISO 25010 Standards.

The defined scope and limitations straightaway enable the goals of this research. The first objective is met by aligning the system at the community level to address the real problems farmers in Buenavista encounter. The second objective is met by integrating the required features including product listing, order management, payment processing, and delivery tracking. The third objective, security and reliability, is achieved through unique account verification, administrator controls, and transparent transaction processes. Finally, the fourth goal is supported by pilot testing, as per ISO 25010 Standards, to ensure that the system’s functionality, effectiveness, and usability are thoroughly tested and verified, even during partial implementation.

1.5 Definition of Terms

The definitions below break down the key ideas and elements discussed while developing and implementing AgriLink, along with some context on how the researchers apply these concepts in their research.

Digitalization – In this study, digitalization refers to the transition from manual or paper-based agricultural market processes to the AgriLink system, which enables farmers and buyers to conduct transactions online.

Digital Platform – The online environment or system that enables users (farmers and buyers) to interact, exchange information, and conduct transactions.

E-Commerce – In this research, e-commerce refers to the internet-based platform (AgriLink) used by farmers and buyers to facilitate transactions, including product listing, order placement, and payment.

Information Systems – In this research, information systems refer to the organized combination of hardware, software, and databases that facilitate AgriLink in storing, processing, and disseminating agricultural information.

Order Management – In this research, order management refers to the systematic process within the AgriLink platform that handles order placement, tracking, confirmation, and completion between farmers and buyers, ensuring smooth and efficient transactions.

Payment System – The digital mechanism that facilitates secure financial transactions between farmers and buyers, including methods like online banking or e-wallets.

Product Listing – In this research, product listing refers to the process of uploading and displaying agricultural products on the AgriLink platform, including details such as product name, quantity, price, and description to help buyers make informed purchasing decisions.

Responsive Website – In this research, a responsive website refers to the design and development approach that enables the AgriLink platform to automatically adjust its layout and features to fit various screen sizes and devices, including smartphones, tablets, and desktop computers. This ensures accessibility and a consistent user experience across all platforms.

Systematization – In this research, systematization refers to the structured process of organizing and integrating various agricultural activities and data within the AgriLink platform to ensure efficiency, accuracy, and ease of access for both farmers and buyers.

System Usability – The ease with which users can operate and interact with the AgriLink platform to perform their intended tasks efficiently.

User Interface (UI) – The visual and interactive part of the AgriLink platform that allows users to navigate, list products, and manage orders efficiently.

User Experience (UX) – In this research, user experience refers to the overall satisfaction and ease of use that users feel when navigating and performing activities on the AgriLink platform.

CHAPTER 2

METHODOLOGY

2.1 Research Design

This project will employ a Design and Development Research (DDR) strategy to develop the AgriLink E-commerce Platform, a solution designed to facilitate the delivery of agricultural products. The DDR strategy ensures that the platform serves farmers and agricultural consumers through repeated cycles of design, development, and evaluation.

2.2 Software Development Life Cycle

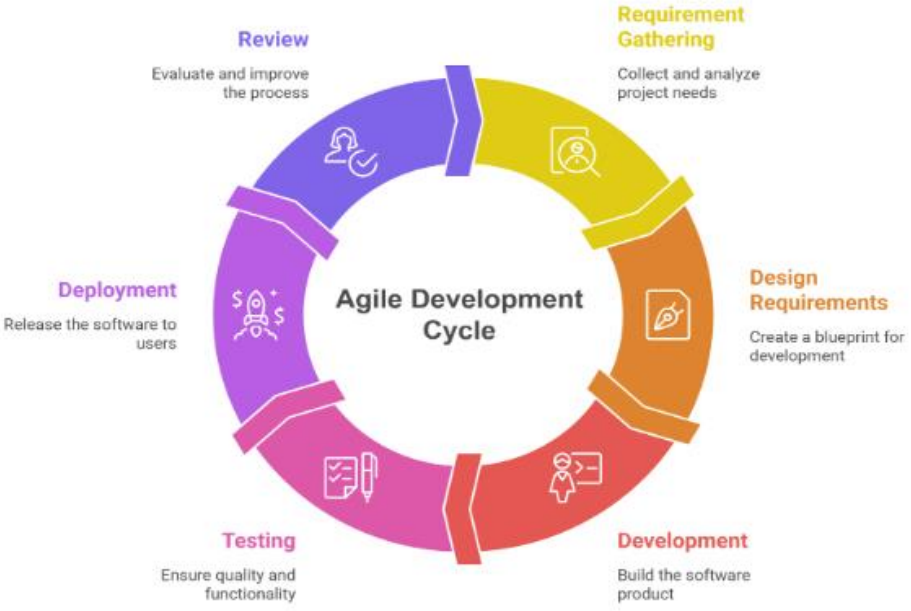


Figure 1. Agile Methodology

The Agile Methodology would be employed in building the AgriLink E-Commerce Platform to ensure it is user-centric, flexible, and responsive to the needs of all stakeholders, including consumers, farmers, and the Department of Agriculture–Buenavista. Agile prefers building the platform in small, functional bites called sprints, which are repeated multiple times. At each level, some component of the operation system will be in the process of implementation by the team, i.e., payment, crop lists, and farmer registration. It is testable and can be improved iteratively before progressing to the next level. The Agile process will also facilitate ongoing fine-tuning based on feedback, allowing for the accommodation of problems such as digital literacy, connectivity constraints, and trust in digital payments. Based on short, timeboxed test-and-decision cycles, AgriLink has the potential to be an inclusive and effective platform that maximizes integration across the farm produce market in Buenavista.

2.3 System Architecture

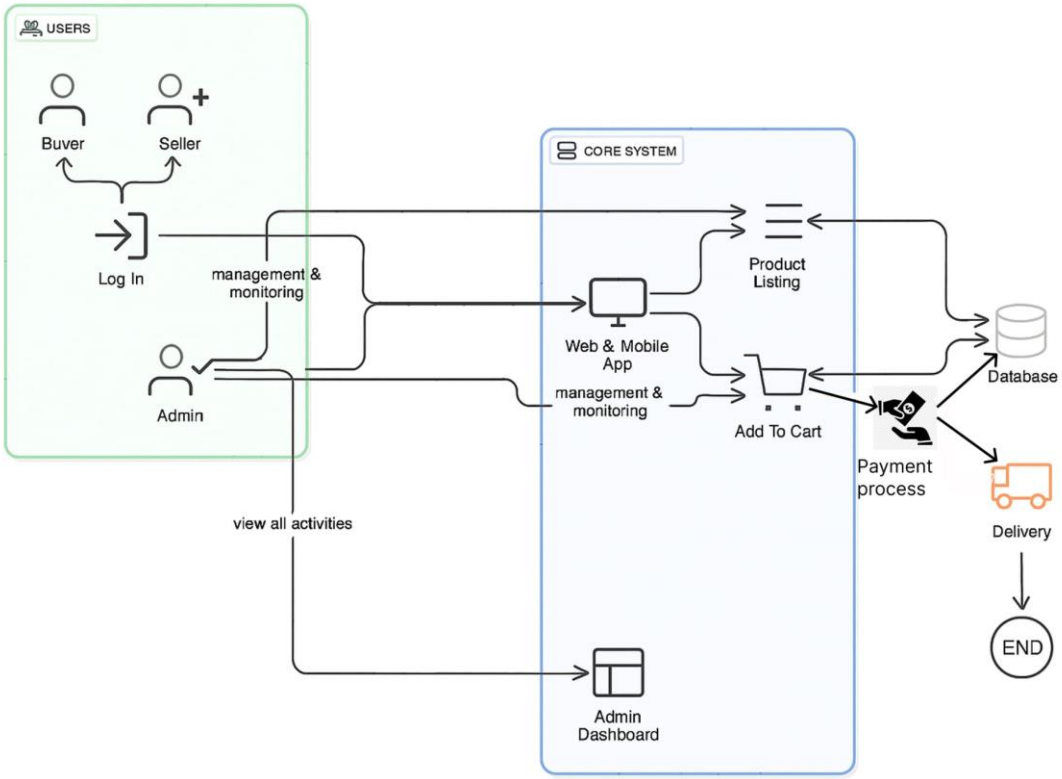


Figure 2. AgriLink (System Architecture)

It demonstrates how the entire AgriLink system functions from start to finish. Individuals who wish to utilize the site, i.e., buyers, sellers, and administrators, log in with the web or a mobile app. The Product Listing module is where sellers add and upload their products. The Add to Cart feature is where buyers search for products and complete their shopping. The Admin Dashboard is where admins manage everything. These modules are linked to the database where user, product, and order data are stored. The database is closely integrated with the core functionalities of the system, including management and monitoring. The database also integrates with the payment gateway area, allowing for secure payments to be processed. Using this arrangement, sellers can present their products, buyers save money by organizing transactions, and administrators can efficiently monitor and manage all transactions.

2.4 Conceptual Framework

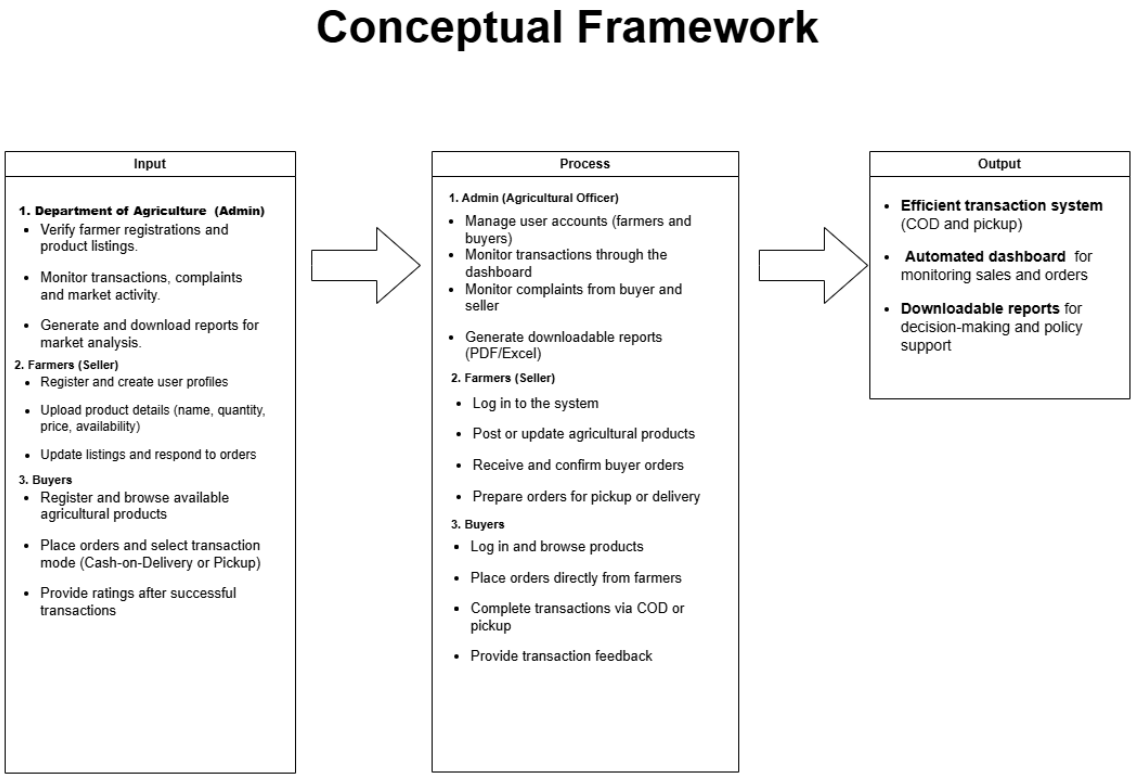


Figure 3. Input-Process-Output Diagram

Figure 3 of the Input-Process-Output (IPO) Diagram of the proposed AgriLink E-Commerce Platform. The Input level has inputs from admins, buyers, and farmers. Farmers input product information, including name, price, quality, and quantity. Buyers input demand information, including queries, orders, and preferences. Meanwhile, admins input admin management for user control and reports. The Process phase illustrates the key processes of the platform, including user login, product management, browsing and searching, cart and order management, payment processing, and administrative management. Lastly, the Output phase illustrates the system's outputs, including direct linkages between farmers and consumers, fair prices, transparent and efficient transactions, empowered communities, and environmentally friendly agricultural trade in Buenavista, Agusan del Norte.

2.5 Use Case Diagram

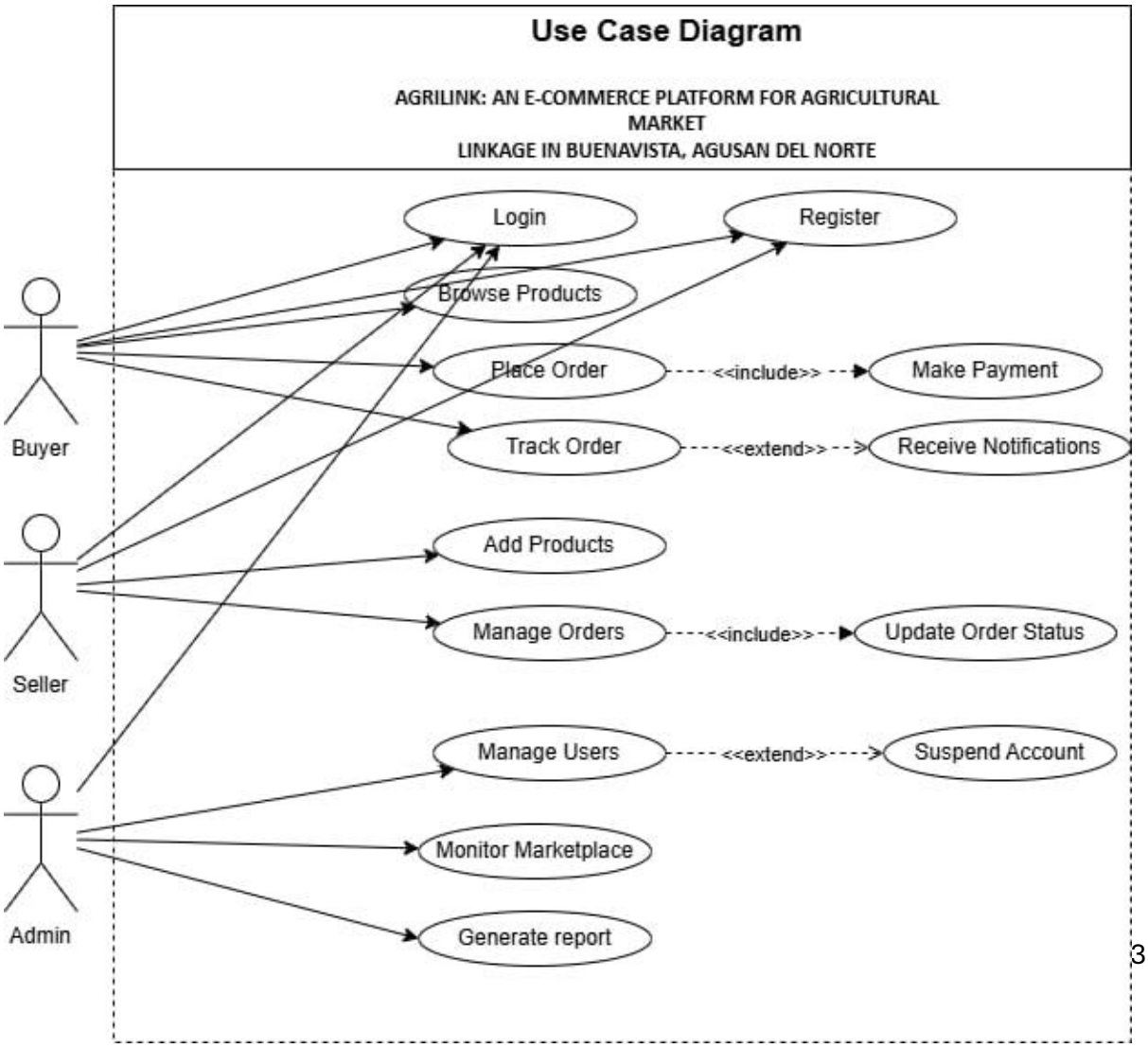


Figure 4. AgriLink Use case

Figure 4 illustrates the interactions among three key actors — Farmers, Buyers, and Administrators — with the system, enabling effective agricultural trade. Farmers can register, update, and delete products, enabling them to manage their products on the platform effectively. Buyers can view available products, add them to their shopping cart, place orders, and make payments, ensuring a seamless buying experience. Admins manage the overall process through handling users, tracking reports, authenticating transactions, and handling conflicts, thereby ensuring that the system remains fair, secure, and transparent. Collectively, these use cases illustrate how AgriLink facilitates farmers in effectively promoting their produce, buyers in making purchases easily, and administrators in keeping the platform running smoothly and reliably.

2.6 Activity Diagram

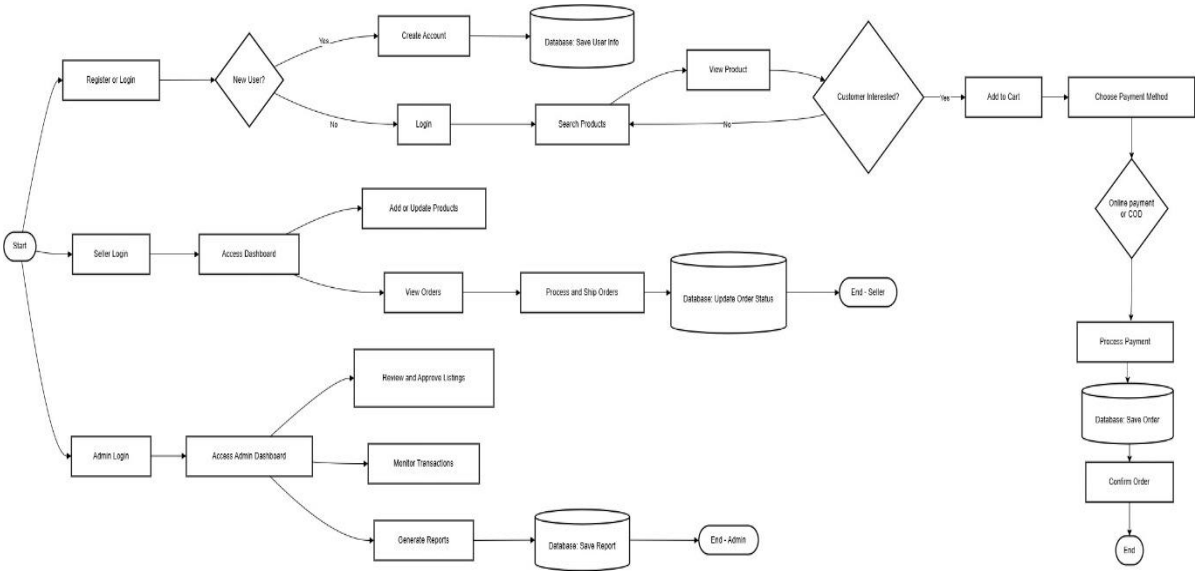


Figure 5. AgriLink Process Flow

Figure 5 illustrates the workflow of the AgriLink E-Commerce Platform, highlighting the activities of farmers, buyers, and administrators within the system. Farmers start by registering or

logging in, verifying their account, and entering their products. These products are then put through for approval, where the admin checks and approves them before they are eventually placed on the platform. Buyers must also register or sign in, verify their account, and continue searching for available products. The system will display the search results, and buyers will be able to add products to their cart, place an order, and complete the payment process. Once the payment is made, the system will validate both the order and the payment, making the process efficient for all parties.

2.7 Sequence Diagram

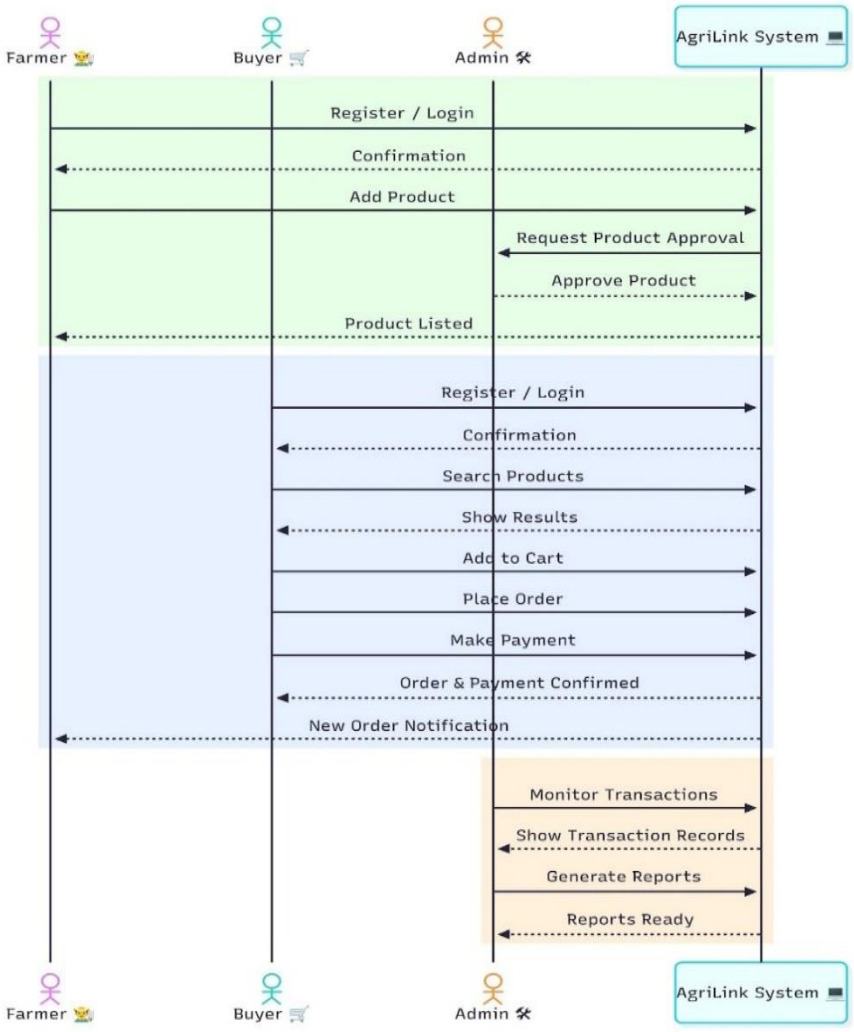


Figure 6. AgriLink Sequence Diagram

Figure 6 presents the Sequence Diagram of the AgriLink E-Commerce Platform, illustrating the sequence of events between buyers, farmers, and administrators. It begins with the registration/login of buyers and farmers. Farmers can add/update their products pending approval by an admin to list them. Buyers, on the other hand, browse/search for existing products, add them to the cart, and proceed to place an order. The customer pays at the time of the order, while the admin handles transactions and generates reports. The order workflow ensures that products are well managed, transactions are verified, and business in the marketplace is evident and optimized.

2.8 ERD (Database Diagram)

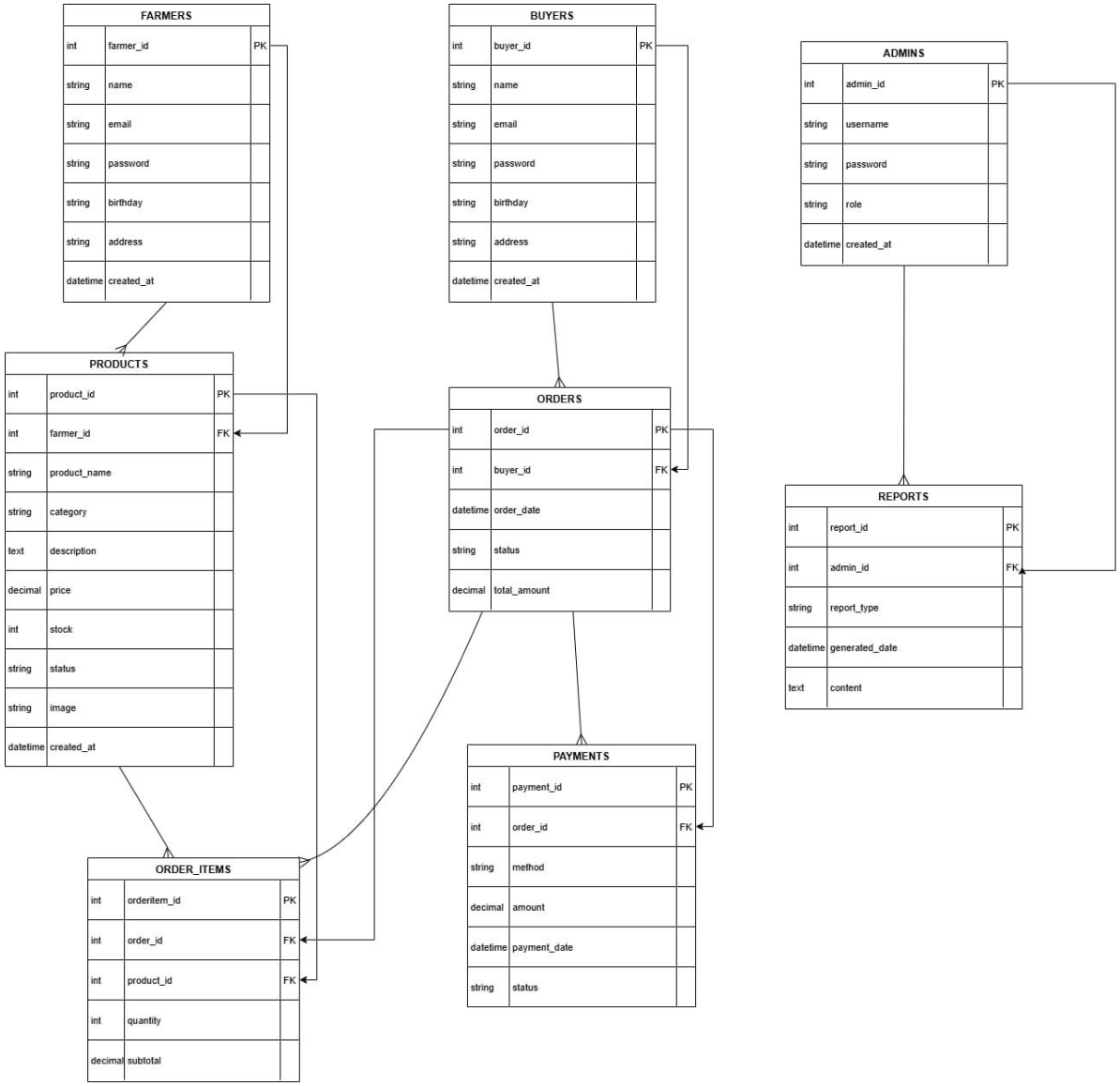


Figure 7. Entity Relationship Diagram

Figure 7 shows the Entity-Relationship Diagram (ERD) of the AgriLink E-Commerce Platform, which depicts the relationships among the system's most important entities. The Farmers entity is related to Products because farmers input and modify agricultural products. The Buyers entity is related to Orders, because orders are purchased within the system. An order contains several Order Items, each representing product information, quantity, and subtotal. Payments are managed by the Payments entity, which defines orders to record transaction methods, amounts, and statuses. The Admins entity generates reports that track activities, transactions, and system performance. This ERD provides a structured database, enabling efficient management of products, orders, payments, and system tracking in AgriLink.

2.9 User Interface Design (Prototype)

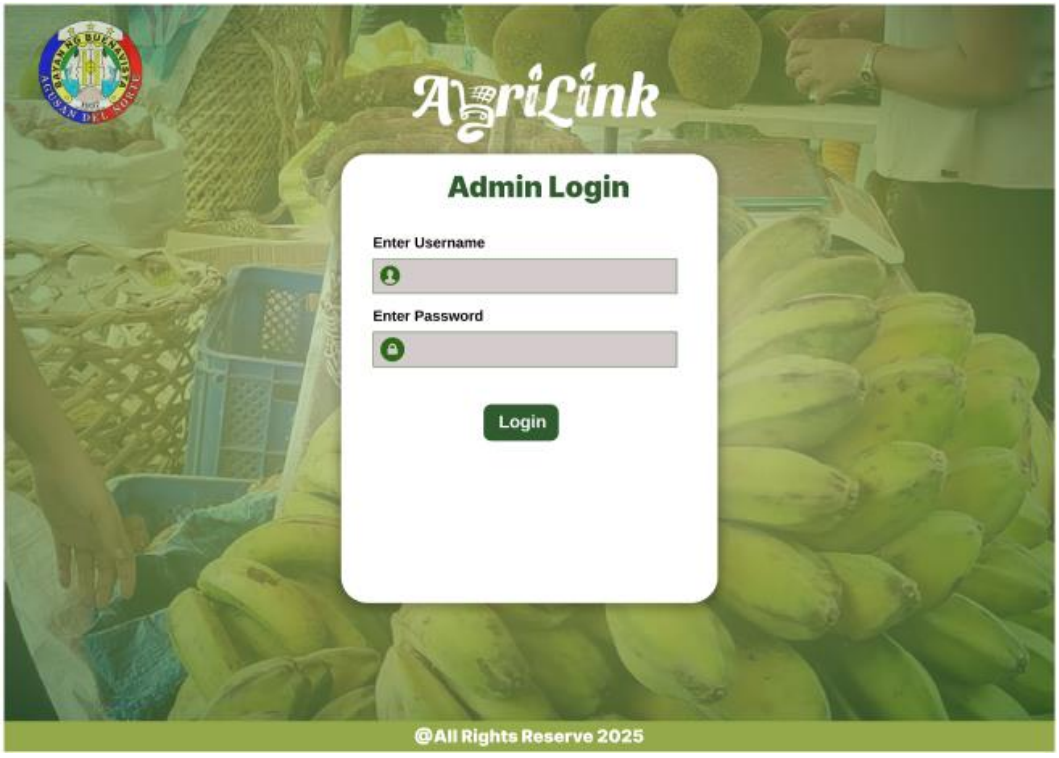


Figure 8. Admin Login

The figure shows a neat, simple admin login page design for the Agrilink Platform. Here are two fields for username and password, with clear labels, and a login button. The layout is designed to be user-friendly, enabling administrators to quickly and securely enter data.

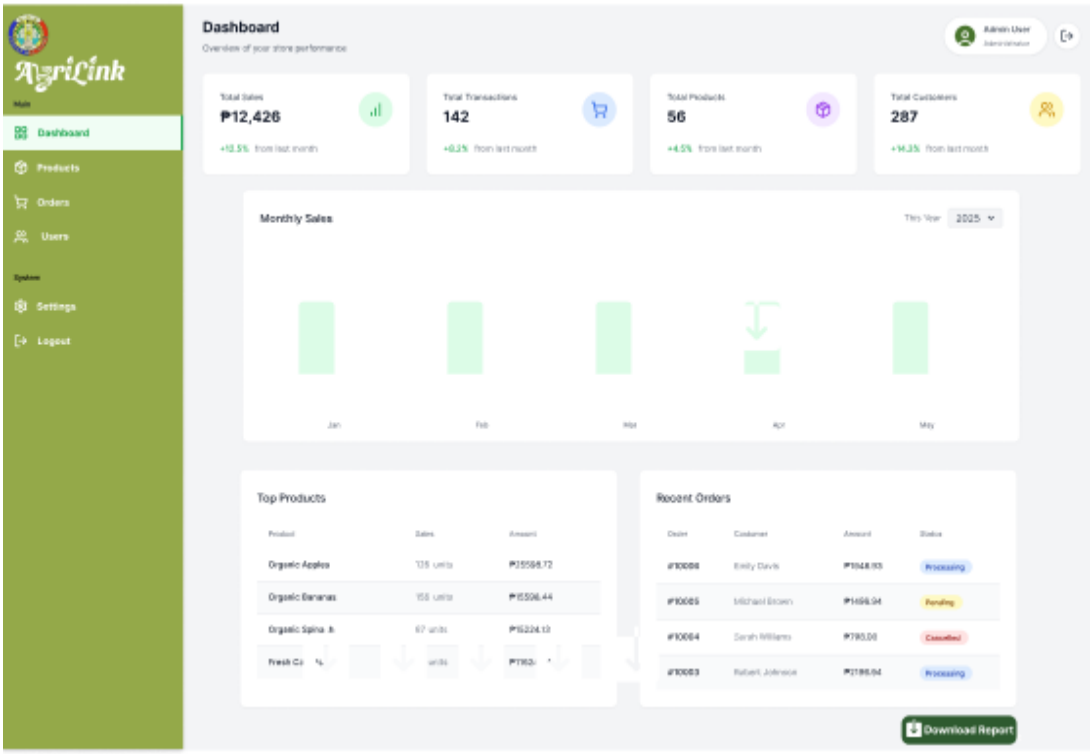


Figure 9. Admin Dashboard

This figure represents the dashboard panel, which provides a comprehensive view of essential business data, including total sales, transactions, products, and customers. The dashboard will feature visual reports on monthly sales, top-selling products, and the latest orders, which facilitate monitoring of the business state. The dashboard enables administrators to monitor sales trends, product popularity, order status, and customer activity. It can also generate reports and provide control over various aspects of the system, improving management.

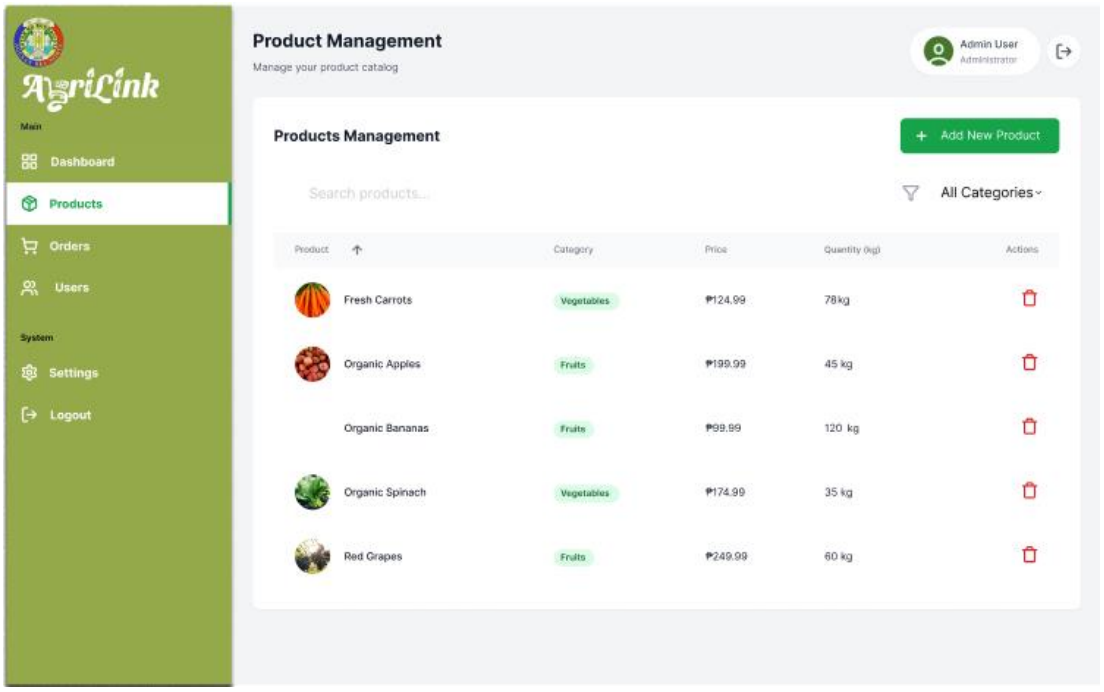


Figure 10. Product Management Dashboard

This figure is Part of the product management on the dashboard allows the admin to manage the catalog of the products. In this section, the admin can see and adjust the list of products, including the name, category (fruits or vegetables), price, and quantity in kilograms. There is also an option to add a new product and delete a current product. It helps to make business products easier and organized.

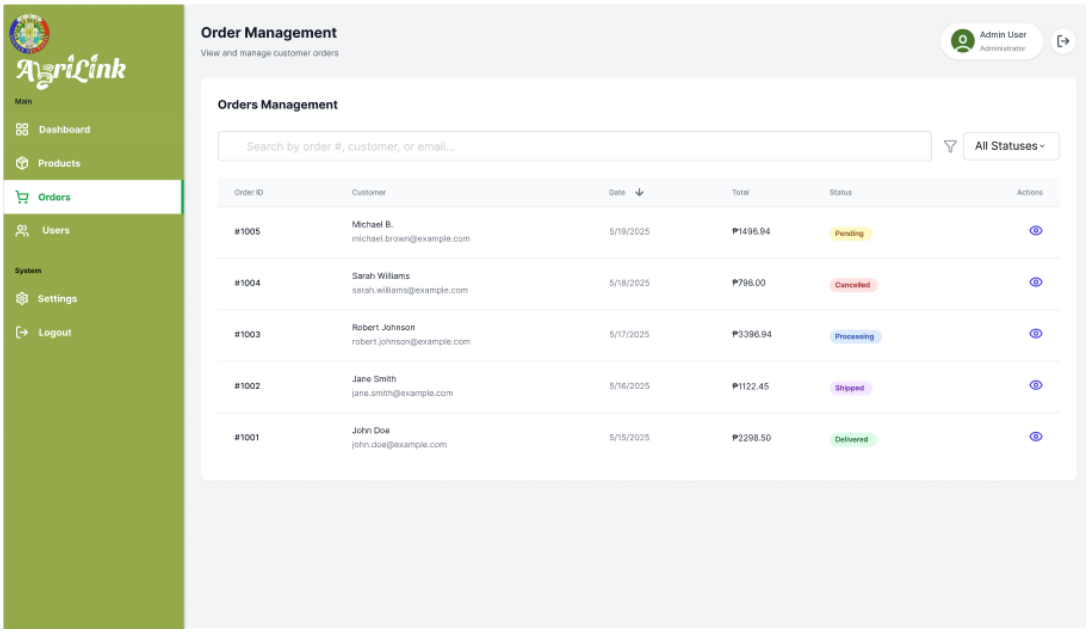


Figure 11. Order Management Section

This figure illustrates the order management section of the dashboard, which enables administrators to efficiently manage customer orders. Here is a list of all orders, including customer name, order date, total amount, and current status, (pending, delivered, or shipping). There is also an option to find a specific order and filter it by status, making it easier to monitor each transaction.

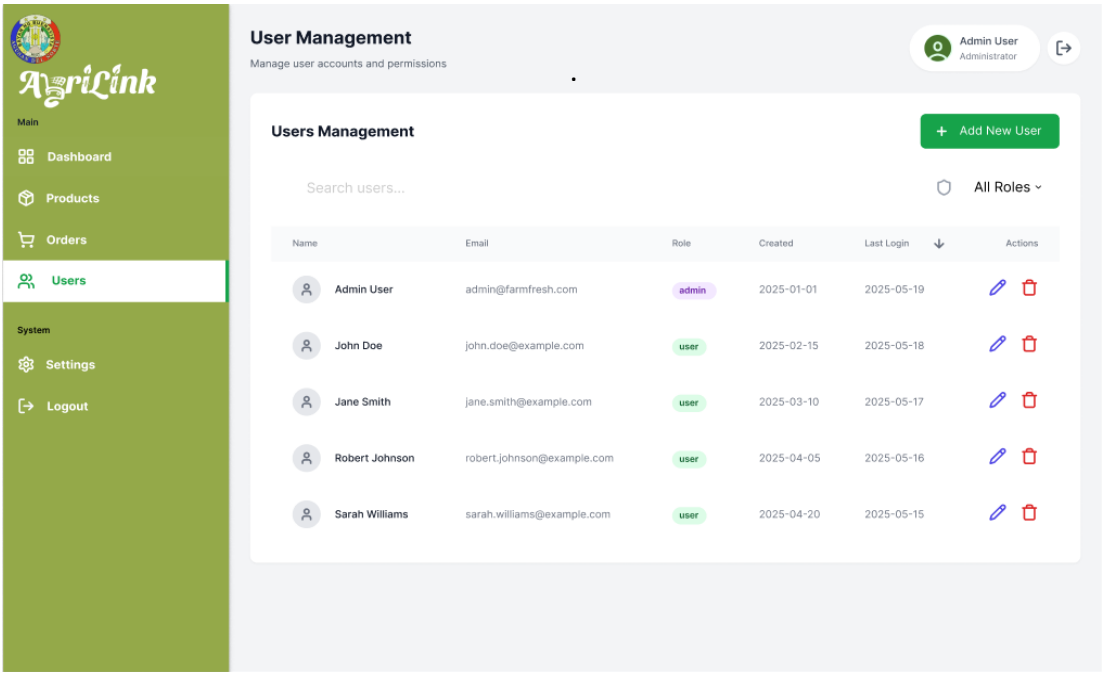


Figure 12. User Management Interface

This figure is the user management interface is designed for easy management of user accounts. Here, the admin can edit and update users' personal information and passwords. The complete list of users will also be found along with their roles and final login. It has a feature to add a new user and search the list of existing users. In this way, the entire process of administration of users in the system is simplified and organized.

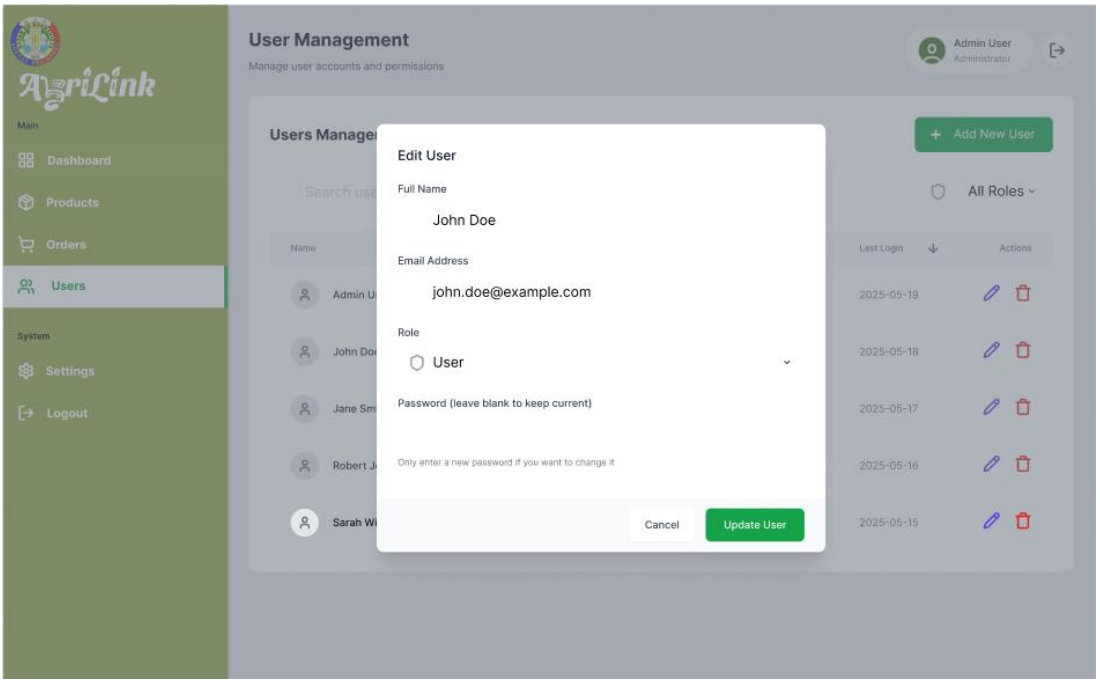


Figure 13. Admin User Management

This figure shows the User Management Page for Admin, where all user accounts on the Agrilink Platform can be managed. The feature to edit User Information, including name, email address, role, and password, is available here. There is also the option to add a new user and view the list of existing users, along with their roles and final login details. On this page, the admin has complete control over managing the permissions and security of the entire system.

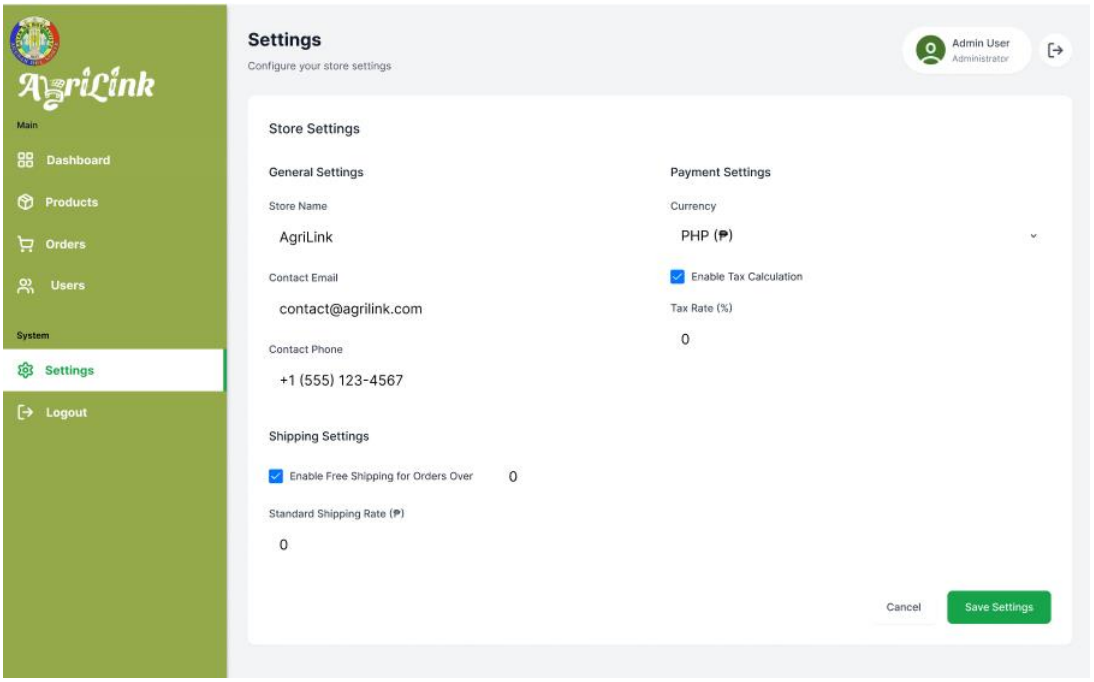


Figure 14. Admin Setting

This figure shows the settings page, a crucial, control panel where the admin can customize and manage key operations of the Agrilink Platform. Here, you can configure general settings, such as platform names and contact details, for communication between farmers and buyers. Shipping settings can also be configured to turn on or off free shipping and set delivery rates. For payment settings, the currency can be selected, and tax calculation can be configured for transactions. Through this page, the admin will ensure that the entire platform for all users can be adequately and effortlessly.

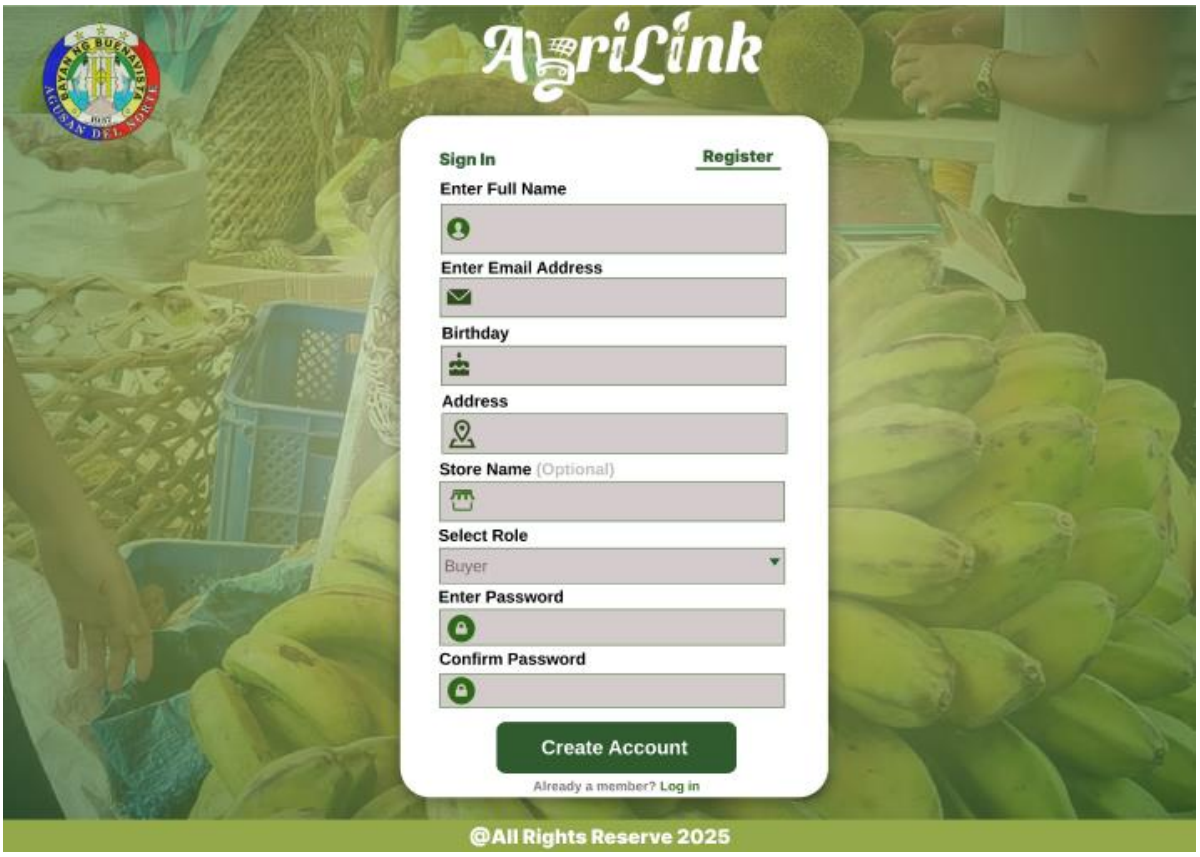


Figure 15. Seller Registration

This figure shows the registration page of the Agrilink Platform where those who want to join a new account can create a new account. The user must fill in the necessary information such as full name, email address, birthday, address, and password. There is an optional field for the store name for those who intend to sell their products. Through this page, farmers and consumers are allowed to be part of Agrilink's digital community, which is the main purpose of our study to cooperate in the agricultural sector through technology.

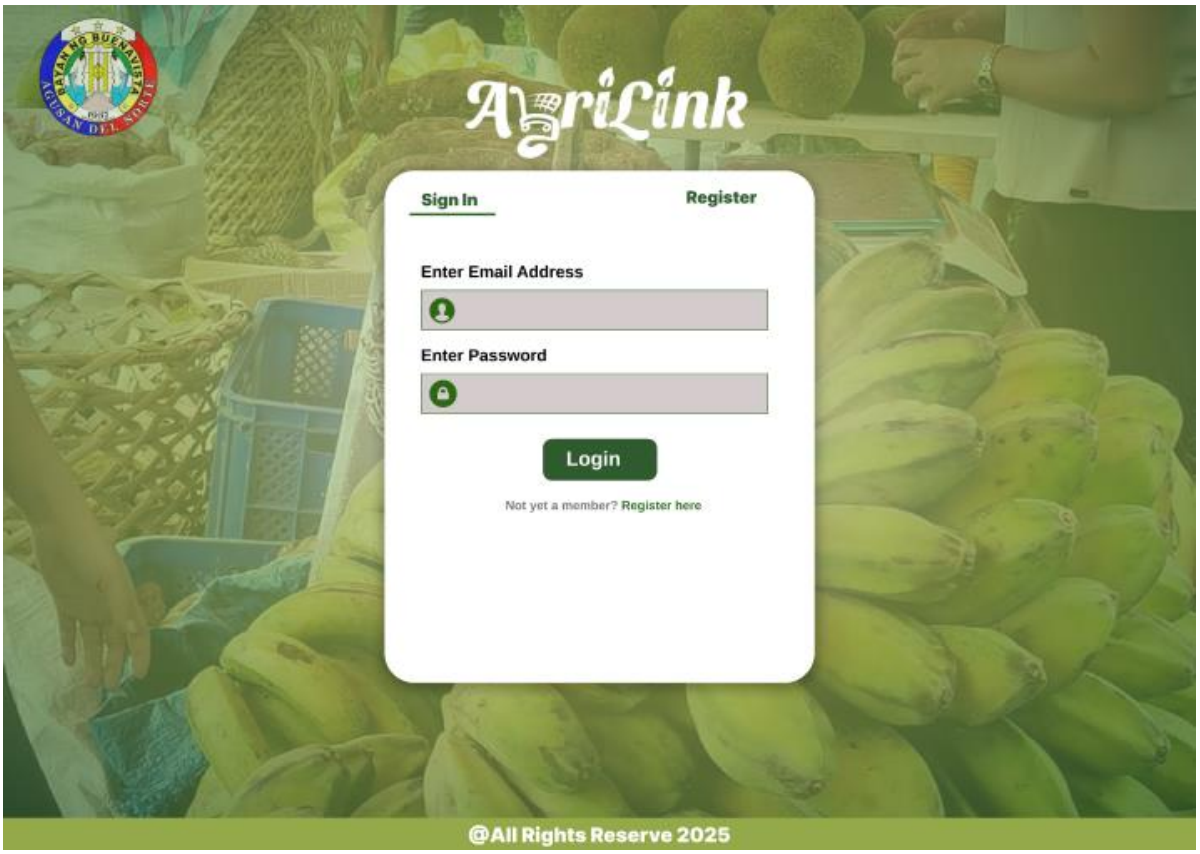


Figure 16. Admin Sign In

This figure shows the sign-in page of the Agrilink Platform, where registered users can log in. Here are two main options: the "Sign In" for existing users and the "Register" for new users who want to join. To log in, the user must place their password in the provided field and press the "Login" button. The simple design of this page reflects our goal to make the platform accessible and user-friendly for all farmers and buyers in Buenavista.



Figure 17. Agrilink's Seller Dashboard

This picture shows Agrilink's Seller Dashboard, where farmers can monitor their sales and demand products. It will feature essential statistics such as the total number of commodities, income, and volume demand for crops. There is also a graphical representation of monthly volume demand to facilitate easier analysis of sales trends. This dashboard supports the study's purpose of empowering farmers through data-driven insights for better planning and management of their agricultural products.

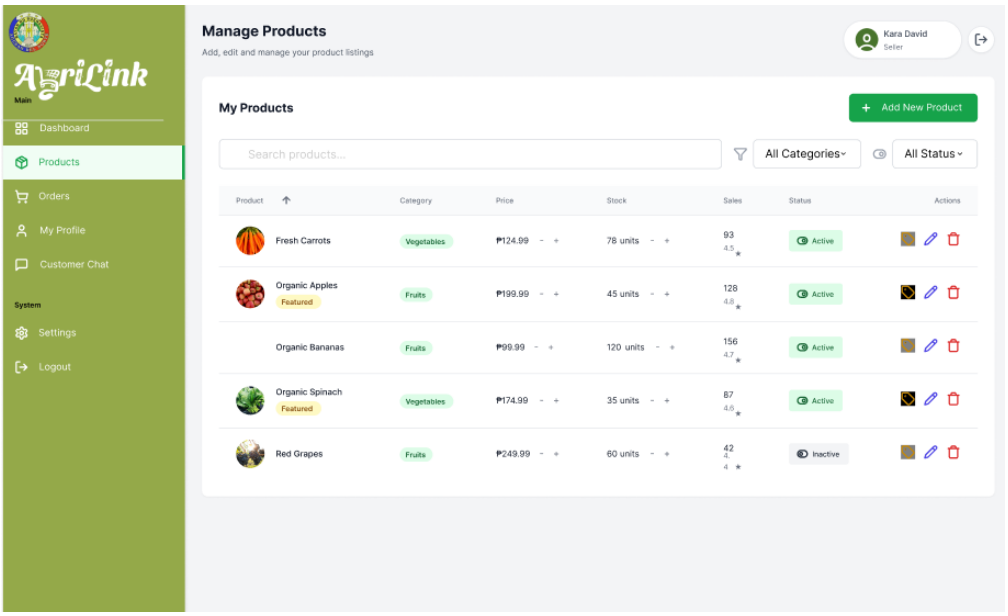


Figure 18. Manage Products

This figure is directly focused on the needs of farmers as sellers on the Agrilink Platform, specifically on the Manage Products page. Here, farmers can manage their agricultural products in an organized way - from adding a new product to monitoring their inventory and sales performance. Through this page, farmers can empower themselves by controlling their online presence, updating prices and stock levels, and tracking the status of each product. This feature supports Agrilink's primary goal of providing direct market access to small farmers and helping them become more competitive in the digital market.

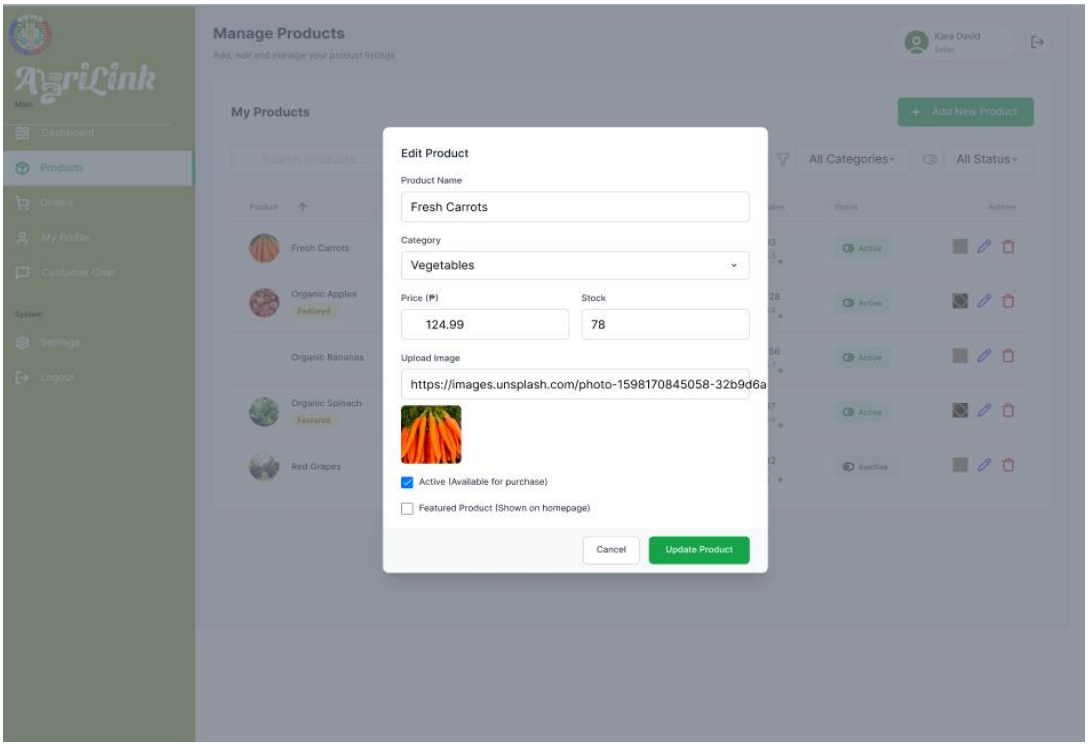


Figure 19. Edit Product

In this picture the "Manage Products" Page will appear, where sellers can edit and update their listed products. There are fields for changing the product name, category, price, and stock. There is also an option to upload a product image and make an item a featured product, which gets more attention on the homepage. Below, you can preview the changes before saving. This page is designed to be user-friendly and straightforward, so farmers can easily manage their online product listings.

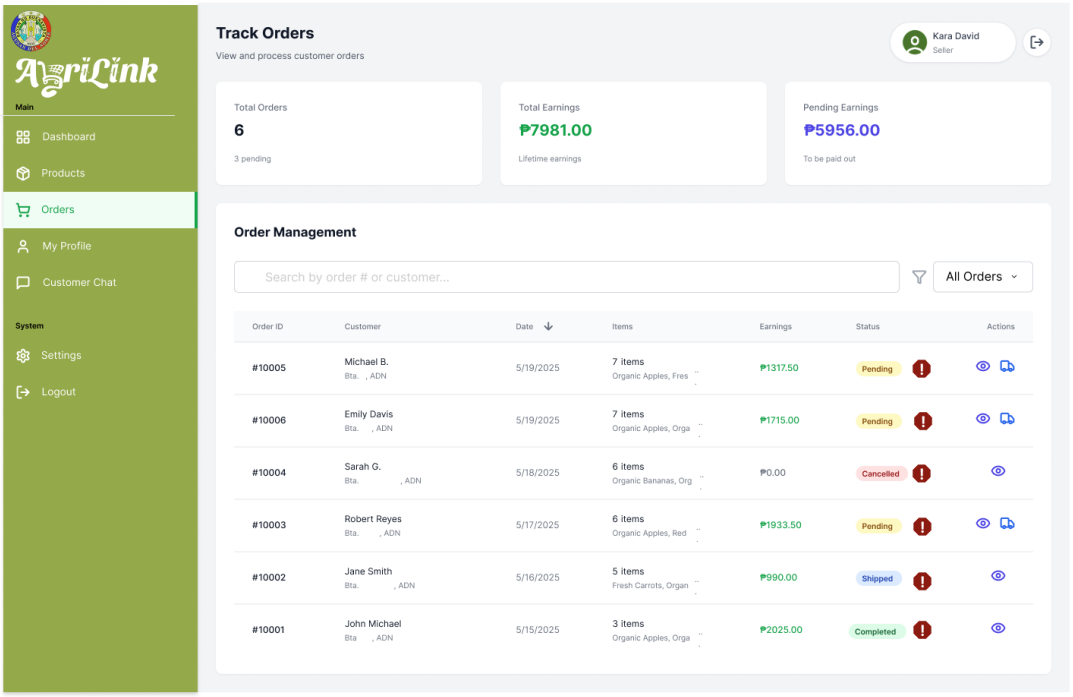


Figure 20. Track Orders

In this picture, the "Track Orders" page is visible, where farmers can monitor and manage all customer orders. It will display a summary of orders, showing a total of six orders and three pending ones. There is also an earnings summary that shows total income and the amount still unpaid.

Below is the list of orders, showing important details such as order ID, customer name, order date, products purchased, total revenue amount, and current status of each order. This page is designed to help farmers easily track their sales and profits in an organized manner. There's also a report button where the seller can report the buyer if they encounter any issue about the buyer.

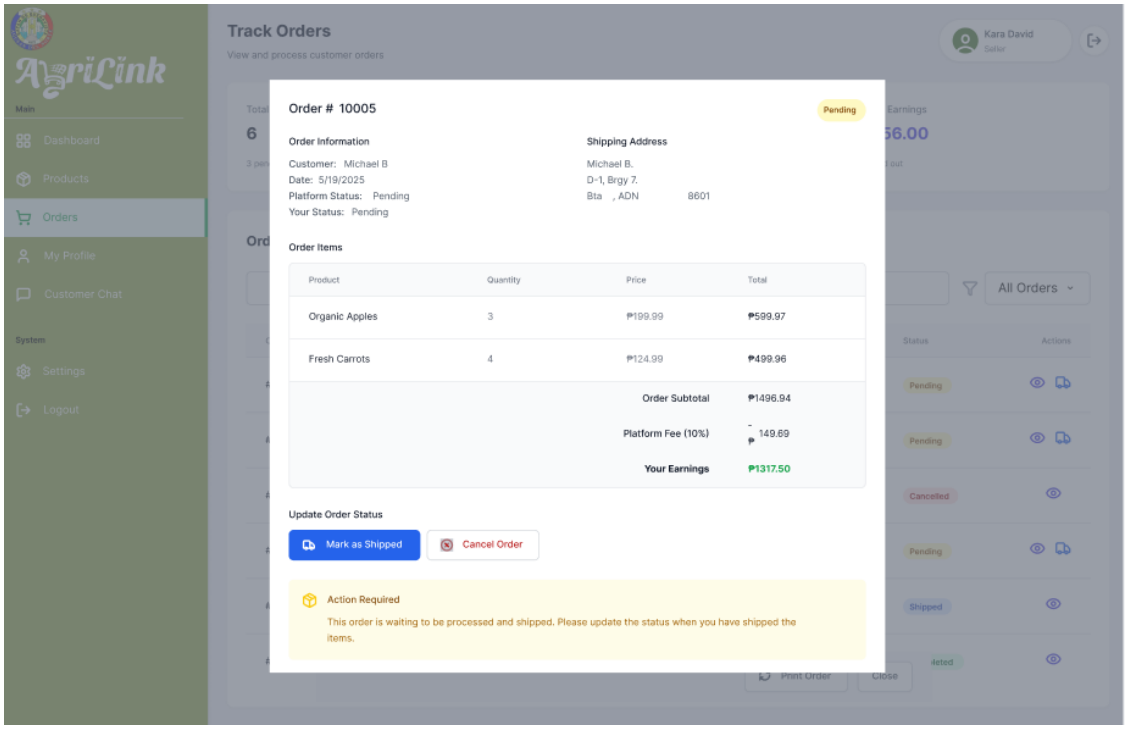


Figure 21. Order Details

This figure provides a detailed view of a specific page, where you can find complete customer information, purchased products, total amounts, and the current order status. There is a breakdown of costs and revenues, including the seller's platform fee and final earnings. There is also an option to update order status and action buttons for order processing. It aims to provide farmers with a comprehensive tool for the effective management of each transaction.

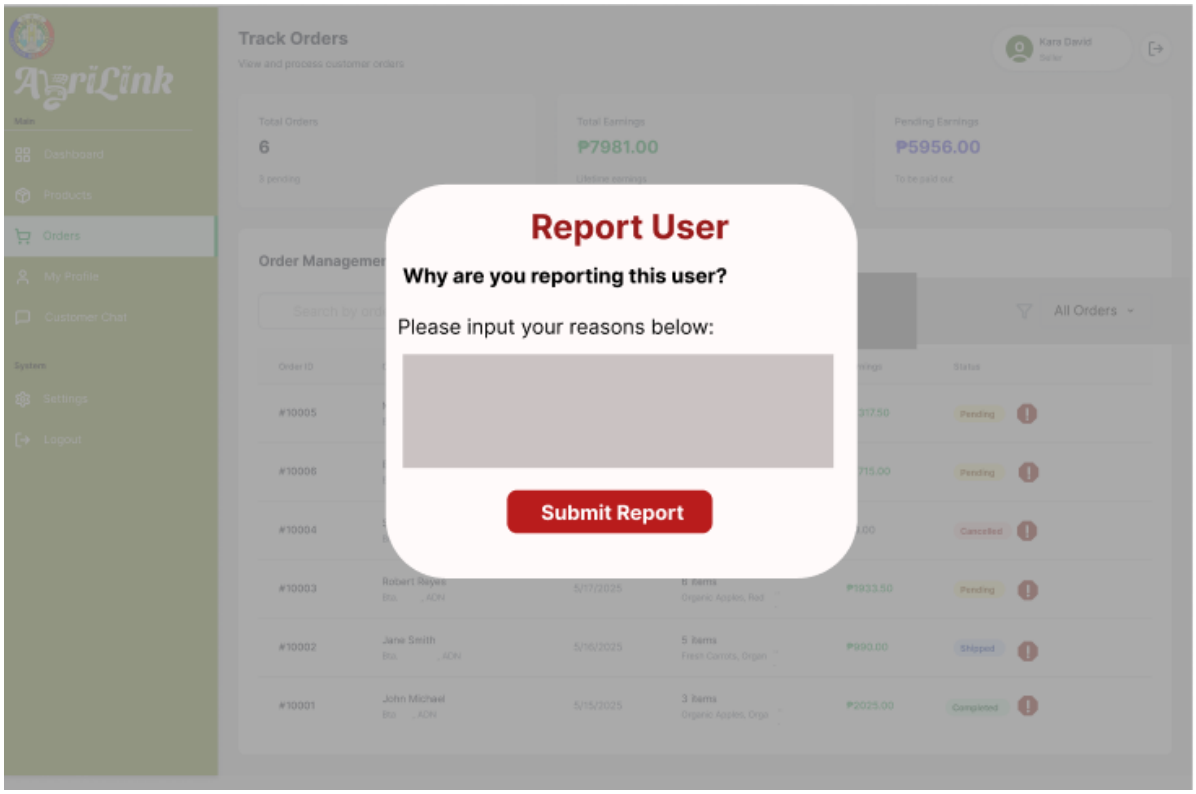


Figure 22. Report User

This figure displays the Report User interface in the AgriLink platform. It enables sellers or buyers to report inappropriate conduct or problems experienced while making transactions. The pop-up form requires the user to identify their reason for reporting another user and offers a text box for detailed input. Upon completion, the report can be submitted for examination by clicking on the Submit Report button. This functionality adds security and credibility to the platform by providing assurance that the issues of users are duly documented and resolved.

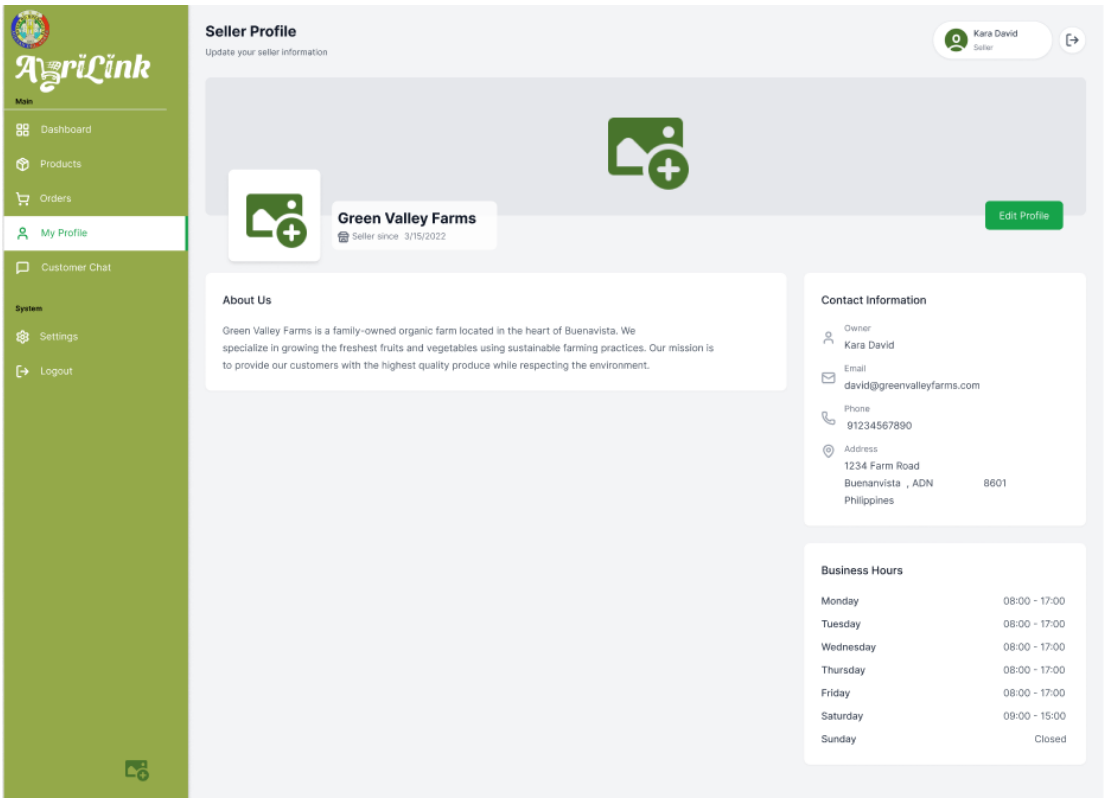


Figure 23. Seller Profile

This figure illustrates the Seller Profile Page, where farmers can view and update their profile information on the Agrilink Platform. This includes the farm's name, a business description, the owner's contact details, and operating hours. Through this page, each farmer is given a personal and professional identity on the platform, which helps build trust and credibility with potential consumers.

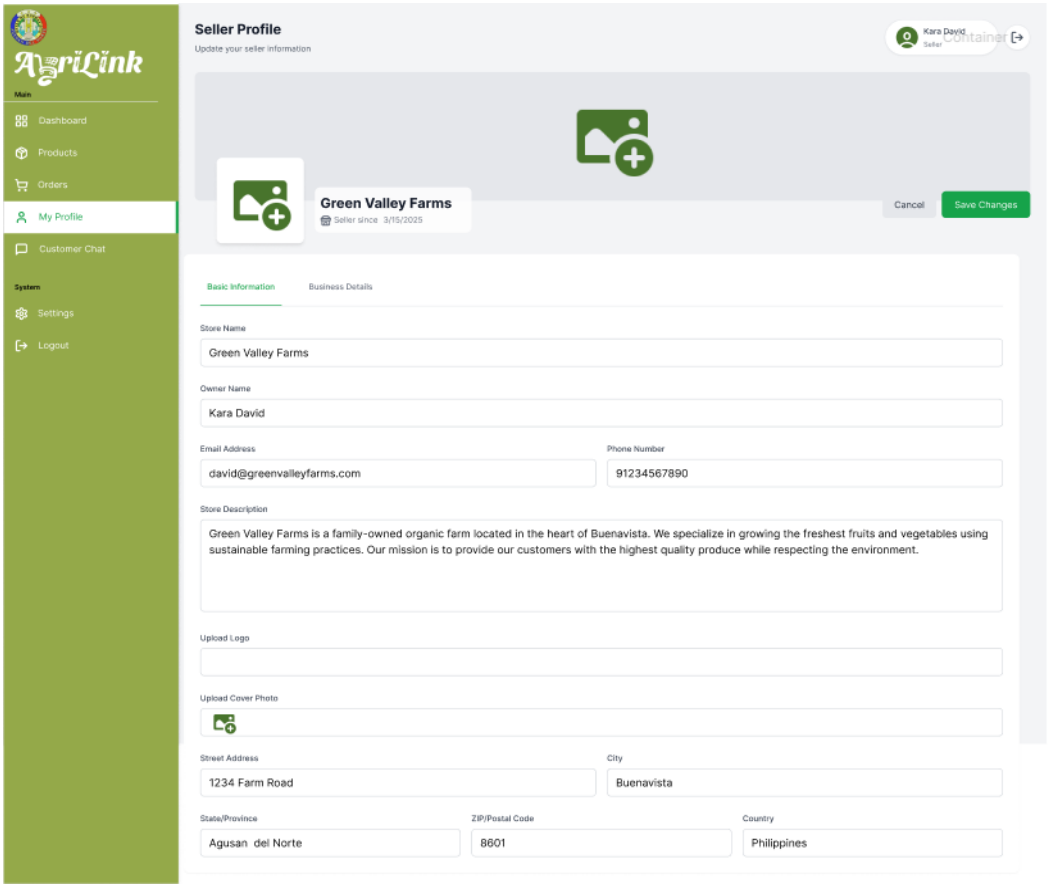


Figure 24. Edit Seller Profile

This figure shows the seller profile page, where farmers can update and manage their information on the Agrilink Platform. Here are the fields for the farm's name, the owner's name, contact details (such as email address and phone number), and a description of their business. Through this page, each farmer is a personal identity on the platform, which is essential to developing confidence and connection with consumers.

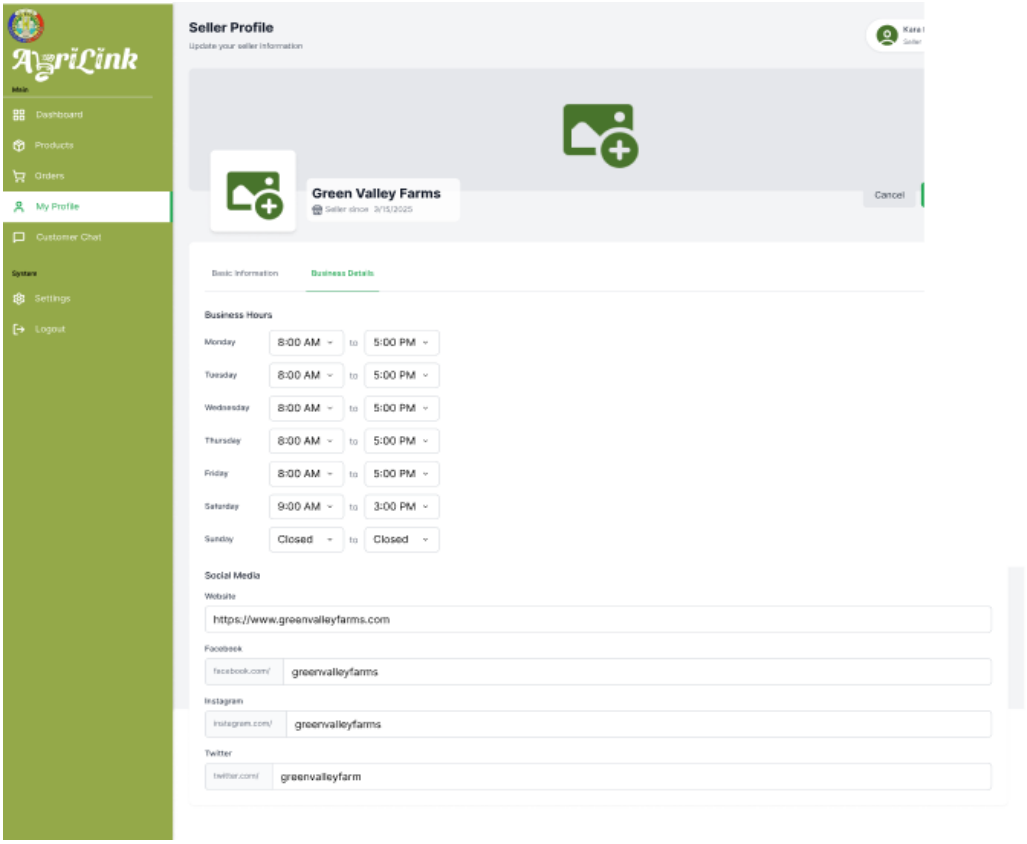


Figure 25. Business Details

This figure shows the seller's profile part, where farmers can set up and update their business hours. There will be a table for each day of the week, from Monday to Saturday, with time slots from 8:00 am to 5:00 pm. This feature is crucial for informing consumers when farmers are available for transactions and communication, thereby facilitating proper coordination between sellers and buyers on the Agrilink Platform.

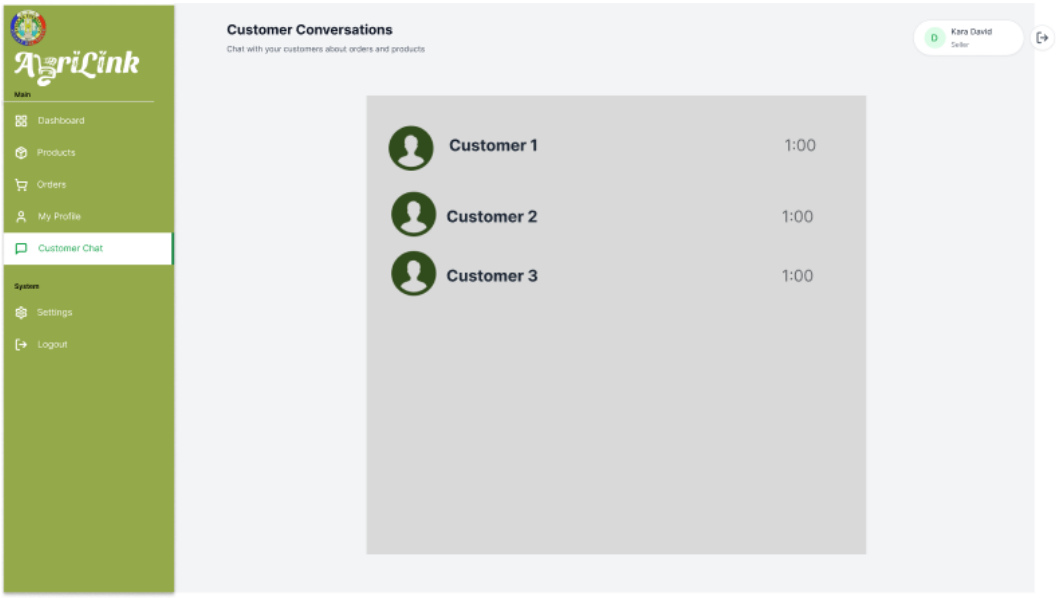


Figure 26. Customer Conversations

This figure shows the Chat Interface where farmers can interact with their customers. Here will be a list of customers with their messages and the time of the last conversation. Through this feature, communication with orders, products, and other transactions is facilitated. Direct chatting enhances confidence and accelerates the business process between farmers and buyers on the Agrilink Platform.

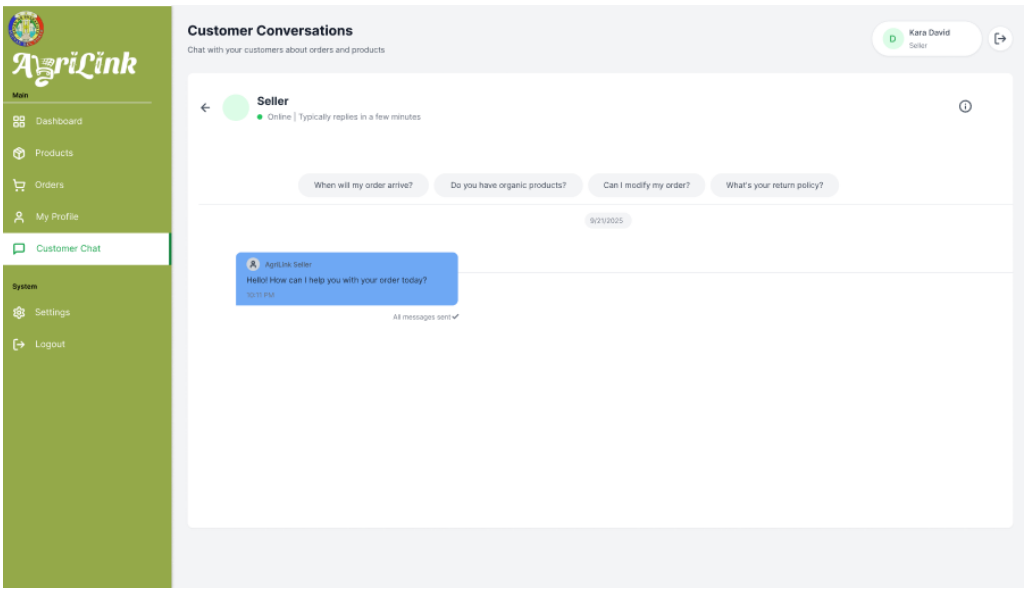


Figure 27. Customer conversation

This figure shows a sales and lead generation page that is Part of the Agrilink Platform. It focuses on collecting information from potential customers through the contact form. Users can enter their email address and phone number in the designated fields. There is a "submit" button below to send details for sales follow-up. This interface aims to facilitate communication between farmers and buyers through effective lead management.

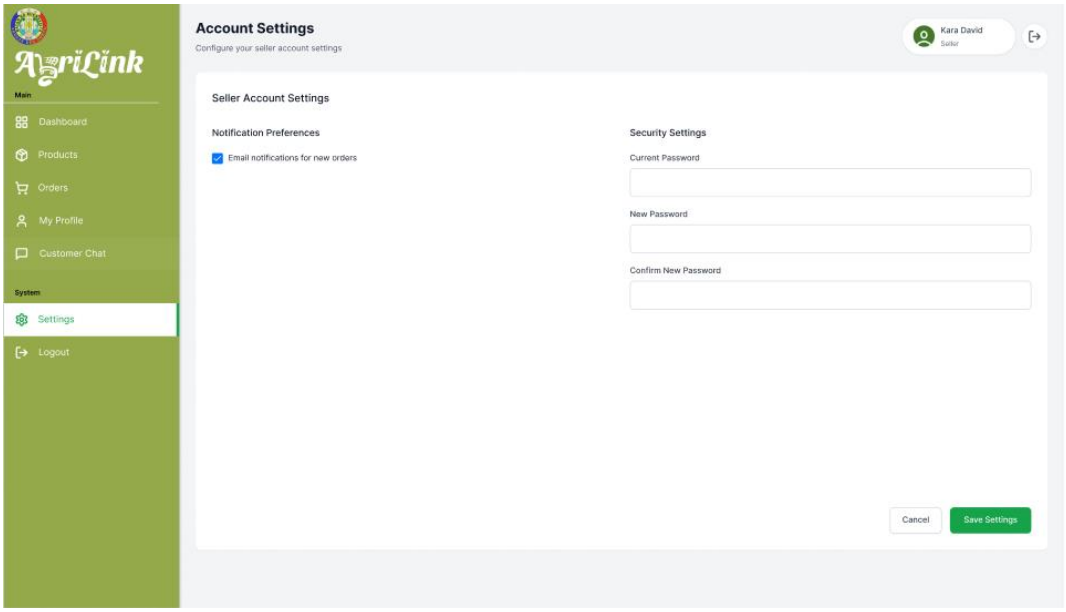


Figure 28. Seller Settings

This figure shows the Account Settings where farmers can customize their account preferences and security settings. Here are the options for notification preferences, such as email notifications for new orders, and security settings for password replacement. Through this page, each user can control their account security and notification preferences, which is essential for a more secure and personalized experience on the Agrilink Platform.

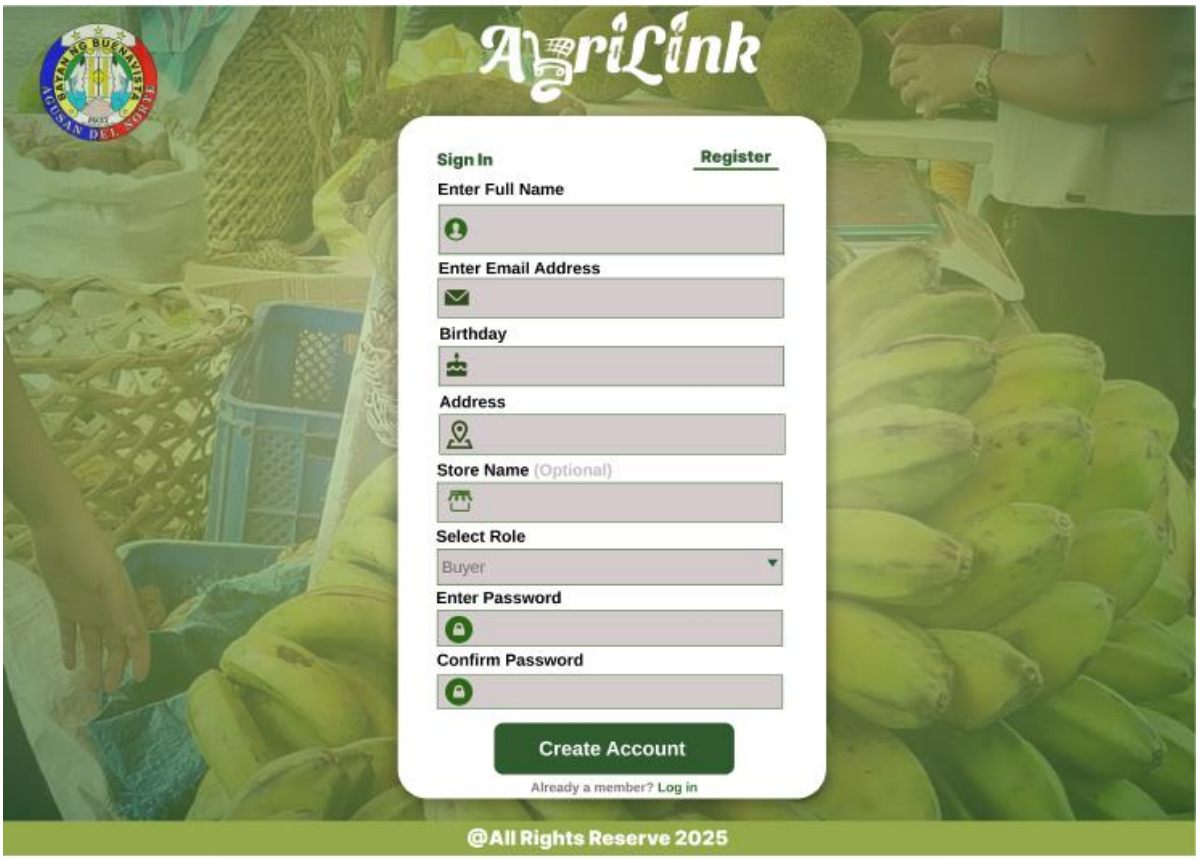


Figure 29. Buyer Registration Form

This figure shows the registration form for buyers on the Agrilink Platform. Here, consumers can register by listing their full name, email address, birthday, address, and creating a password. There is an optional field for the store name, which may also be used by buyers who intend to become future sellers. Through this form, consumers are given access to the platform to help grow local agriculture by direct purchases from Buenavista farmers.

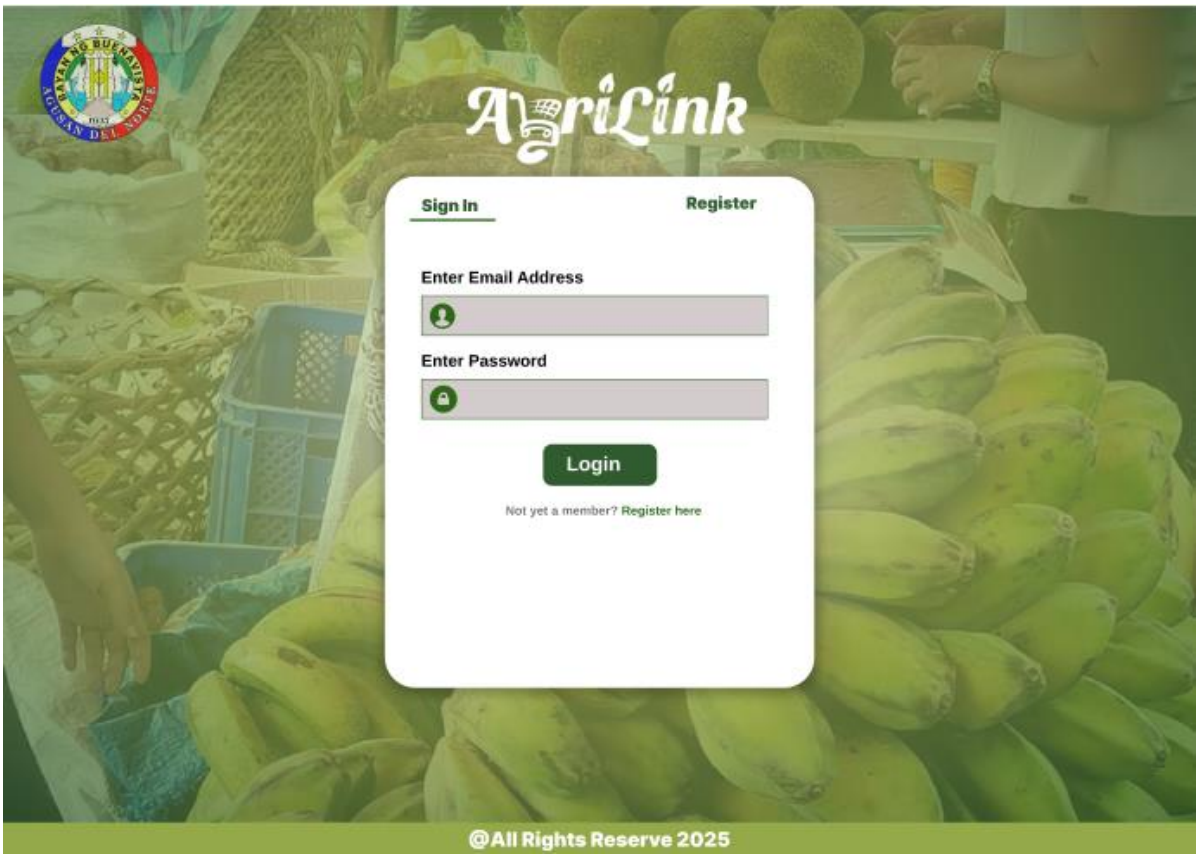


Figure 30. Buyers Sign In

This figure shows the Sign-in Page for buyers who have already registered on the Agrilink Platform. Here, they can now log in by entering their password and pressing the "Login" button. This page serves as a gateway for consumers to access their accounts and explore and purchase fresh products from Buenavista farmers.

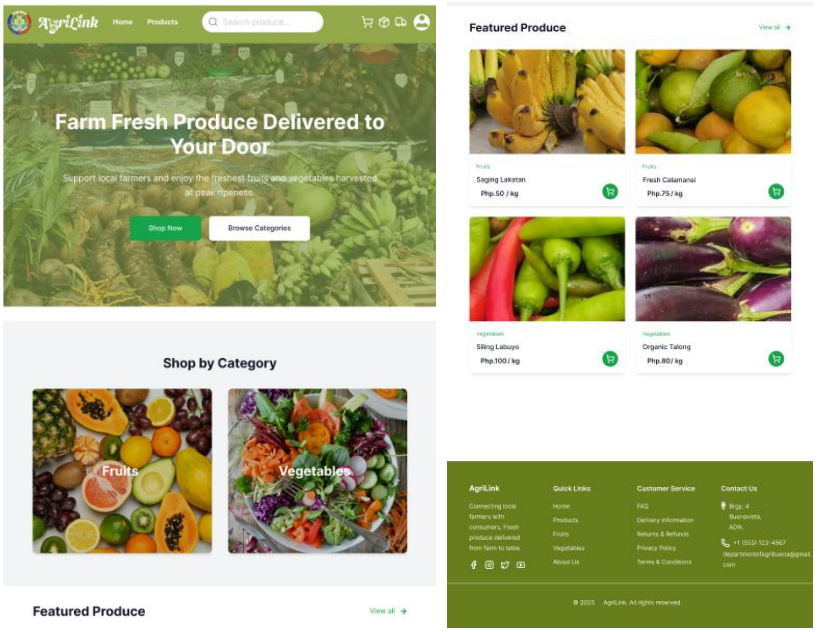


Figure 31. Agrilink Consumer Homepage

This figure is the homepage of the Agrilink Platform, designed for consumers. It can see the primary purpose of the platform: to deliver fresh produce from local farmers to the consumer's door. There is a feature to shop by category, such as fruits and vegetables, and a section for featured produce that showcases products like avocados, carrots, and organic tomatoes. Through this homepage, the primary purpose of Agrilink is to establish a direct relationship between farmers and consumers.

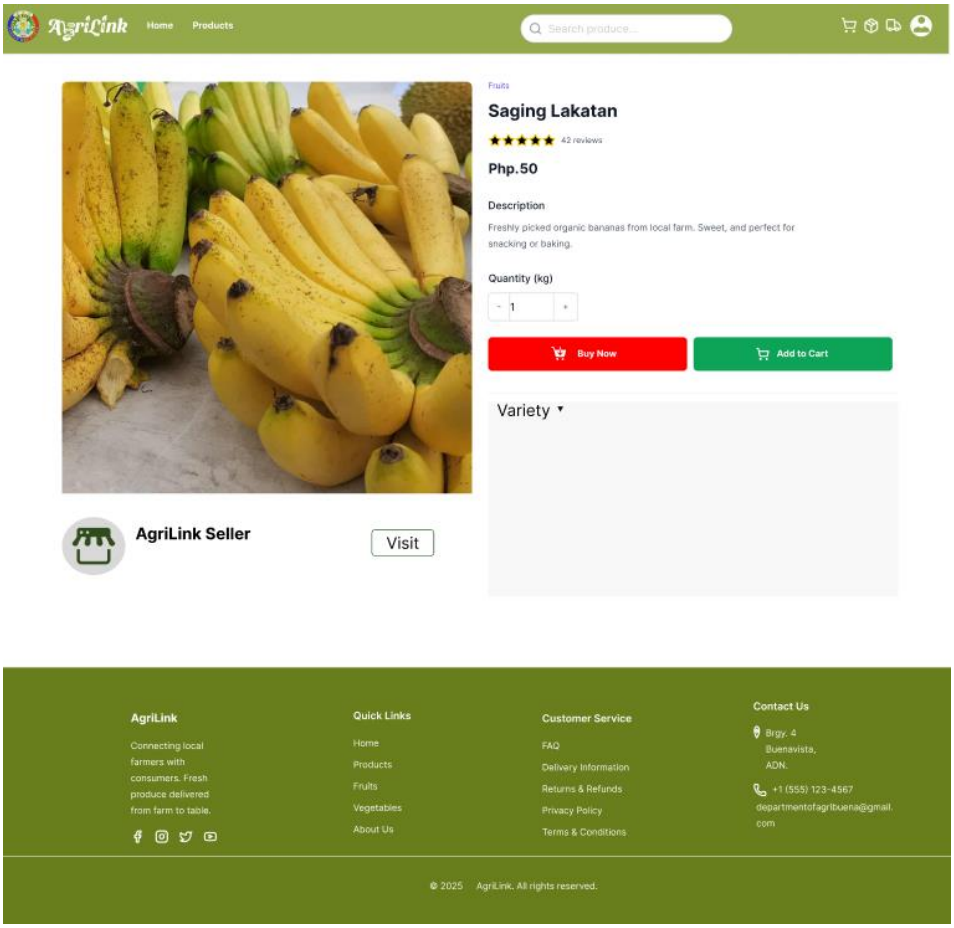


Figure 32. Product Details

This figure shows the product listing page where consumers can view detailed information about a product. It will find the name of the product (banana), price (Php 50), and description of the quality and source of the product. There is also a quantity selector, allowing the buyer to choose the desired amount. Below, the seller's name will be displayed, along with the option to visit their store. This page aims to provide comprehensive information to consumers, enabling them to make informed purchase decisions.

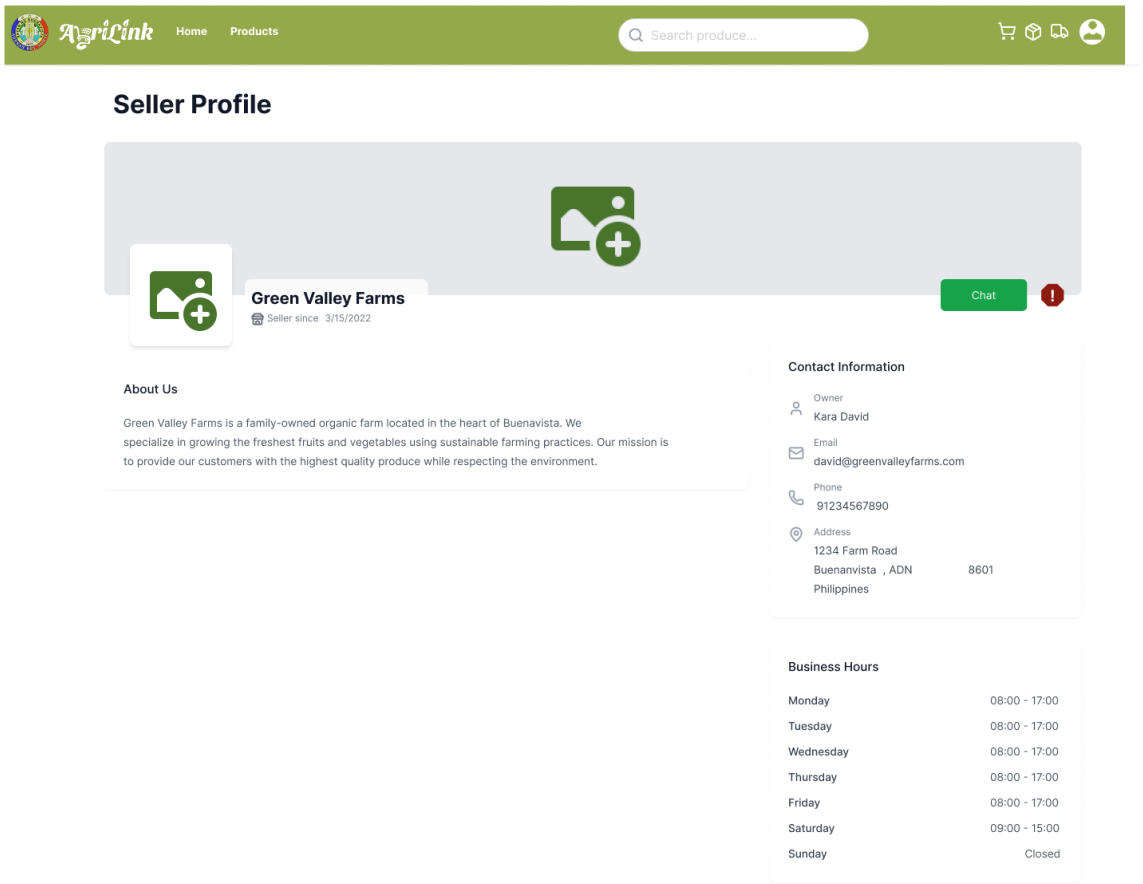


Figure 33. Seller Profile (Consumer View)

This is the Seller Profile Page, in which consumers will be able to check the profiles of the farmers in Agrilink. It will offer the name of the farm, its business profile, contact number, and working hours. With this page, consumers will be able to relate to the farmers who produce the goods they consume, thus deepening trust and higher transparency in every transaction. There's also a report button wherein buyers can report the seller's profile if they encounter any problem with the seller.

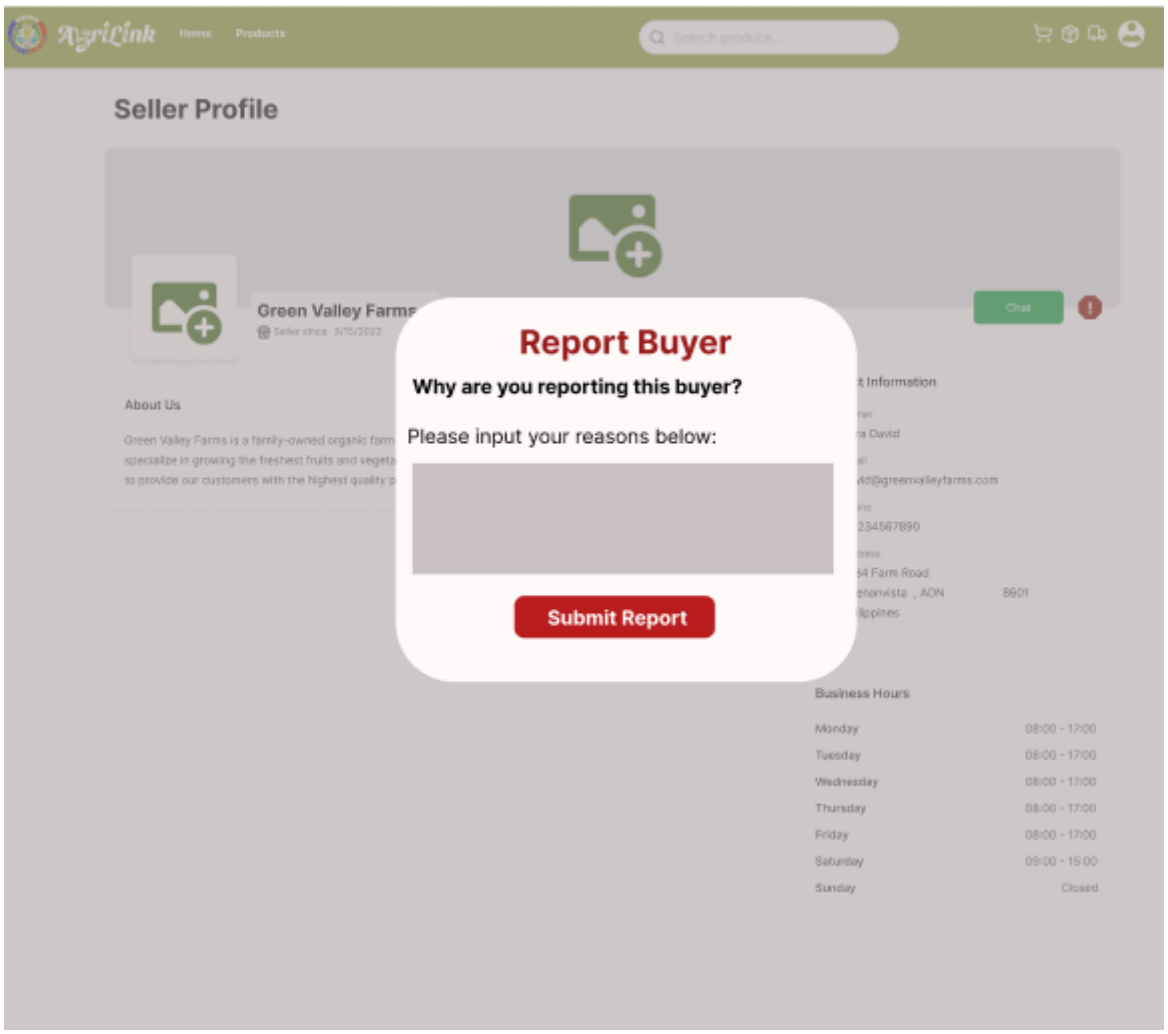


Figure 34. Report Buyer

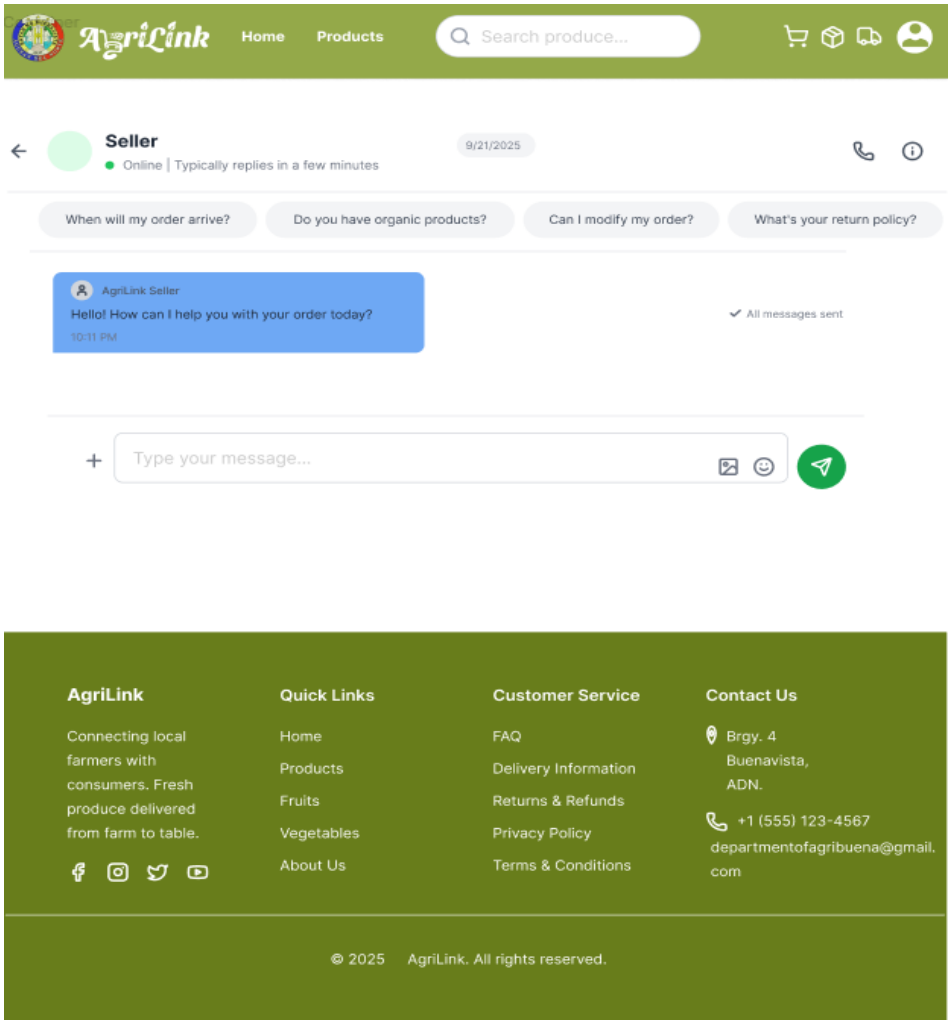


Figure 35. Buyer–Seller Chat Interface

This is the Chat Interface, through which consumers can directly talk with farmers. It will be able to view the seller's "online" status and their average response time. Below is a message box to insert a personal message. This feature will enhance communication and trust between buyers and farmers on the Agrilink Platform.

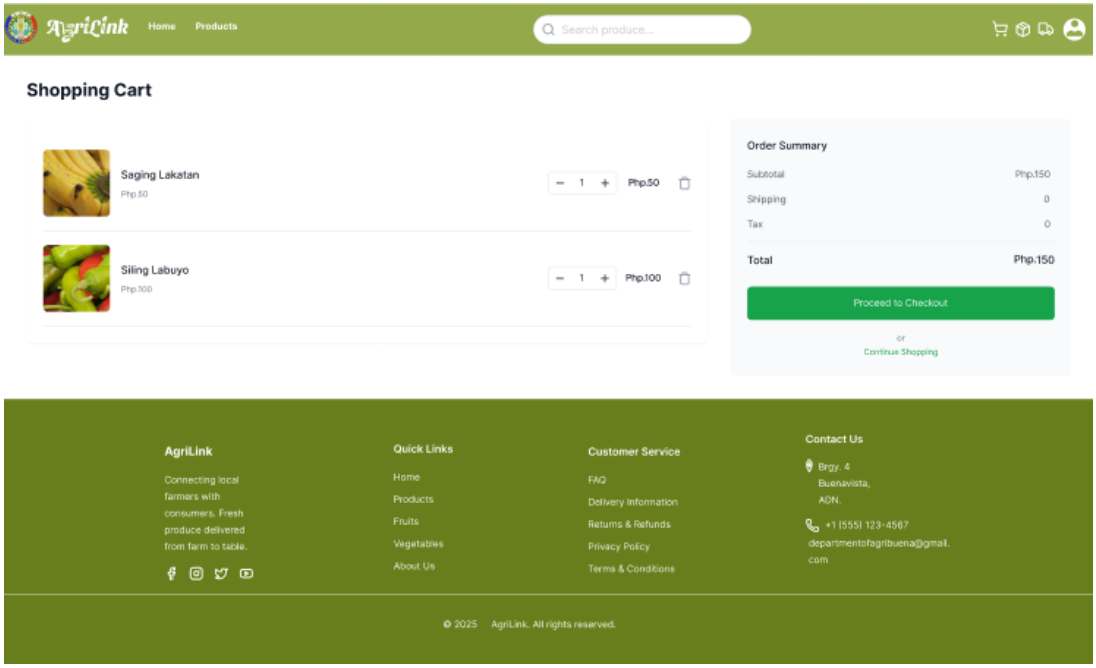


Figure 36. Shopping Cart

On this page, you have the option to see the complete amount of their order before payment. This is the Shopping Cart Page, where customers see their choices before they proceed to pay. Below is a list of products placed in the cart along with their prices and amounts. There is an order summary showing the subtotal, shipping charges, and total payment amount. Buyers coordinate their desire to buy through this page.

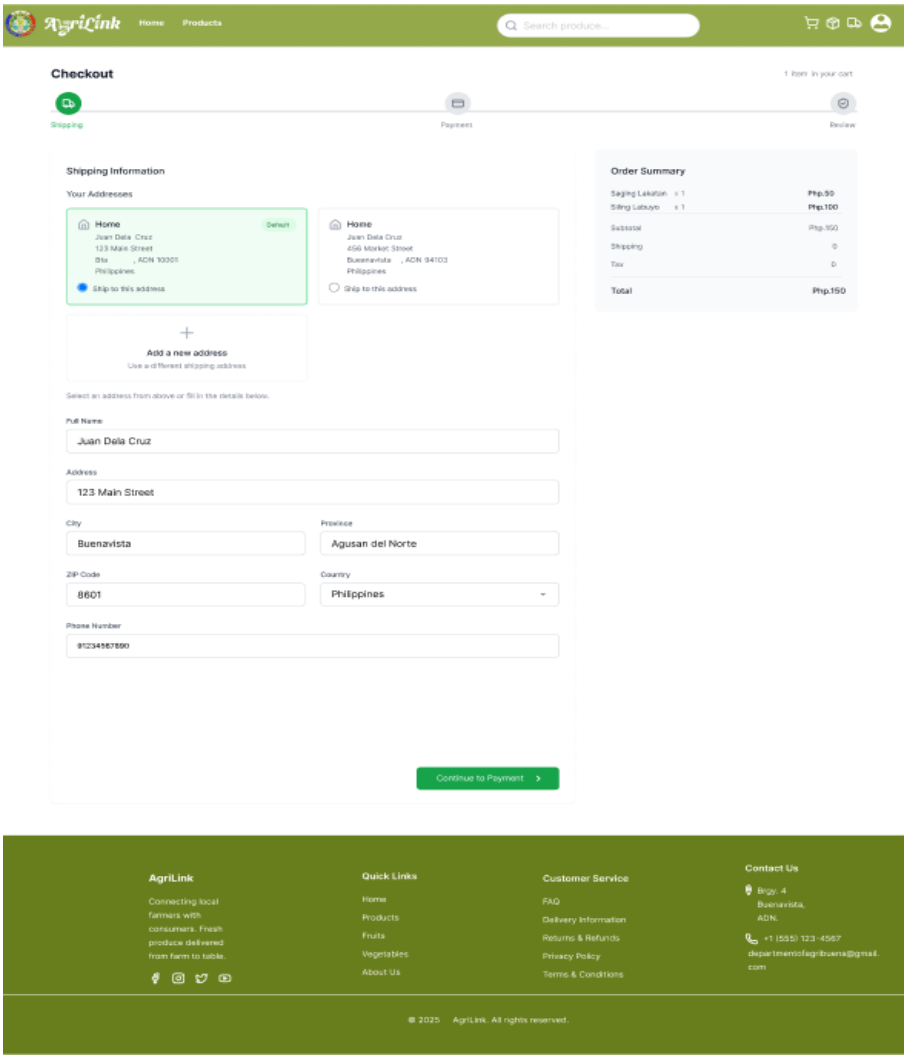


Figure 37. Checkout Page

The page is the payment checkout page on which customers can make their purchase. Included on this page are the stored addresses that the customer can select for shipping, as well as a summary order with the products, totals, and shipping costs. There is also an option to add a new address if necessary. With this page, customers enjoy a hassle-free checkout process before the final payment step.

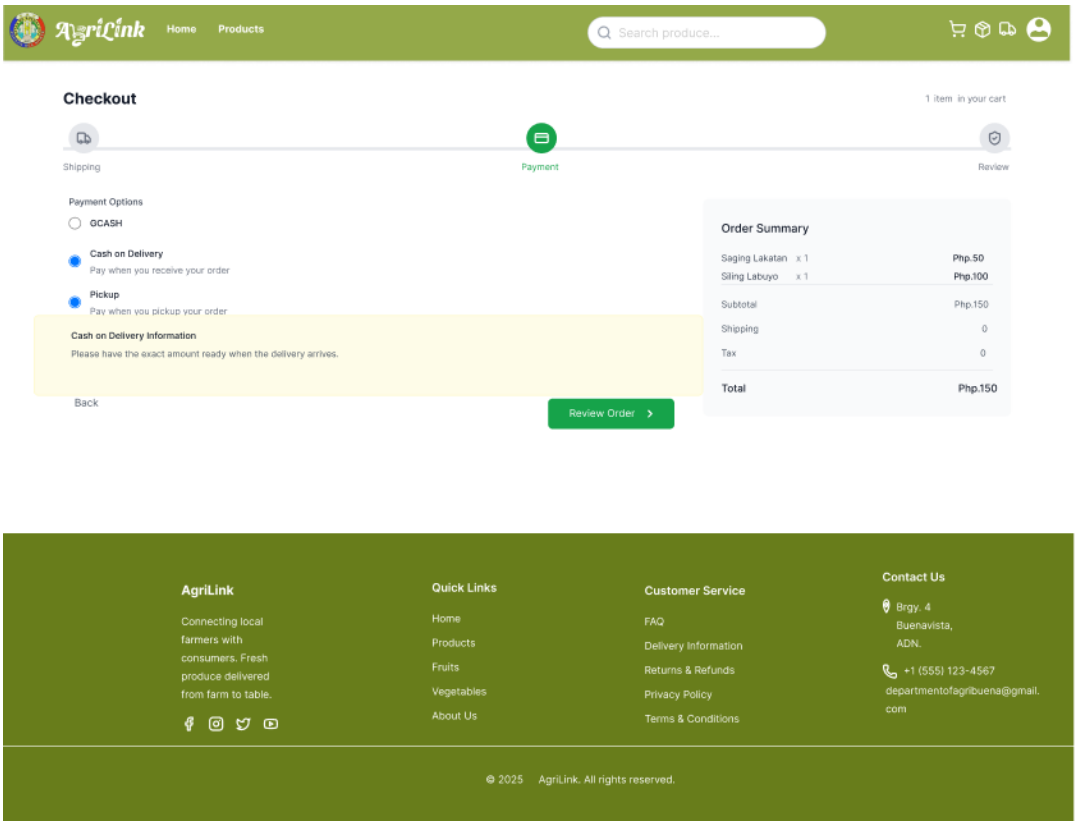


Figure 38. Payment Selection

This is the Payment Selection Page, in which customers are allowed to select their preferred payment option. There are options like GCash, Cash on Delivery, and Pickup. There are details for all the available payments, especially for cash-on-delivery, where the exact amount should be paid at delivery time. It also shows the summary of the order, including the amount purchased and the total amount. Through this website, customers can choose the payment mode that suits them best, providing flexibility and convenience.

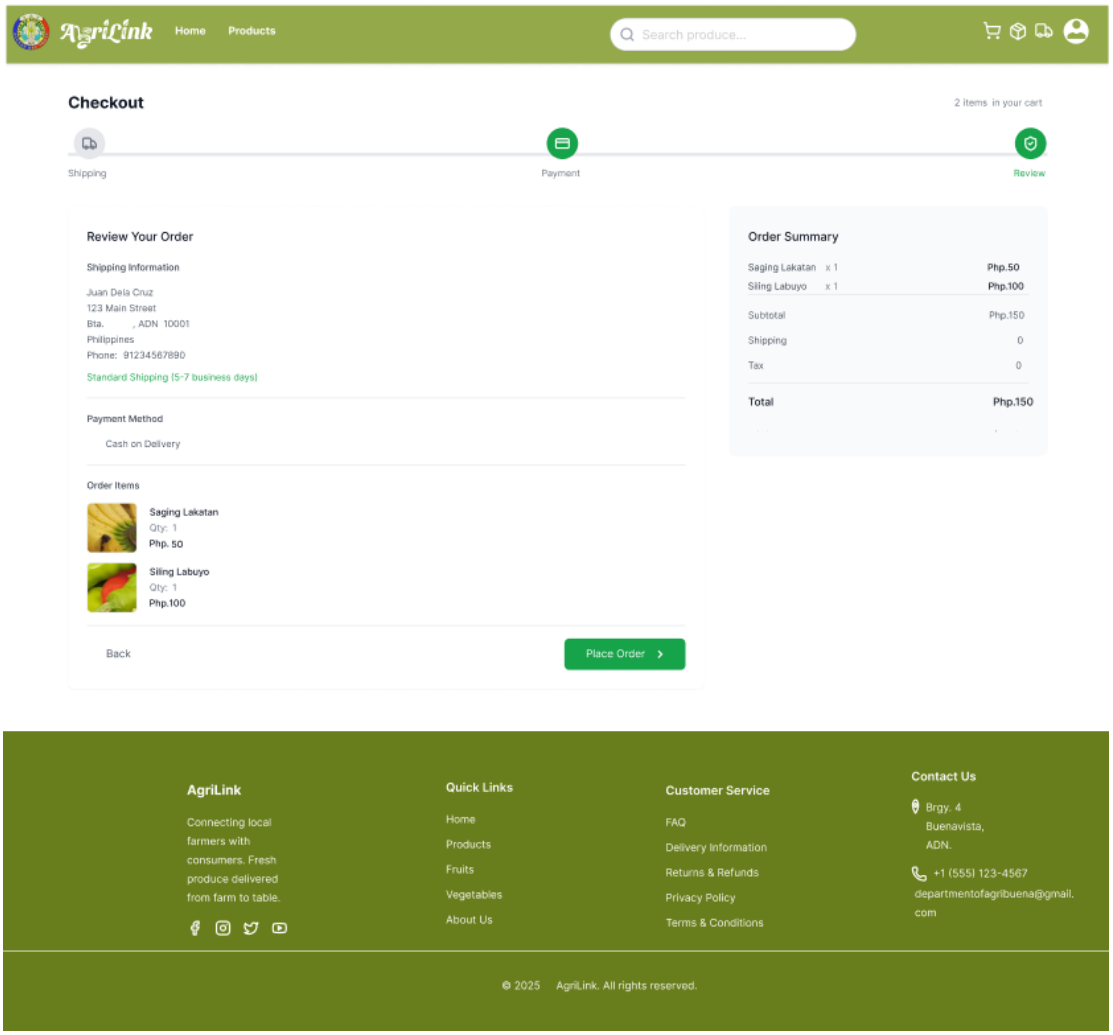


Figure 39. Order Review

This page is the Order Review Page, where customers can verify their order before it is processed. It will provide complete information, including shipping address, payment method (cash on delivery), and a list of items purchased. A summary order with the total price of the purchase is also provided. On this page, the purchaser can view and verify all the information in their order before finalizing it.

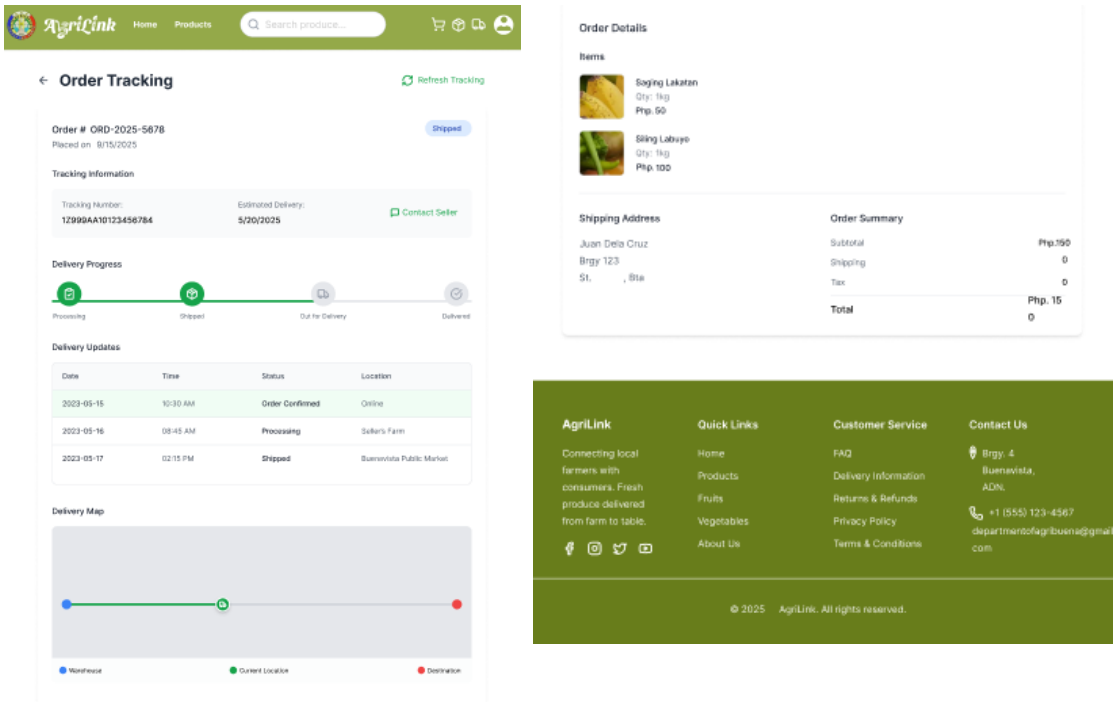


Figure 40. Order Tracking

This is the order tracking page, where customers can see the status of their order. You can view the order number, order date, tracking number, and delivery progress updates here, which reflect the status from order confirmation to delivery. There are also full order details, including products purchased, shipping addresses, and an order summary. Customers are offered transparency and reassurance through real-time information on the status of their orders via this page.

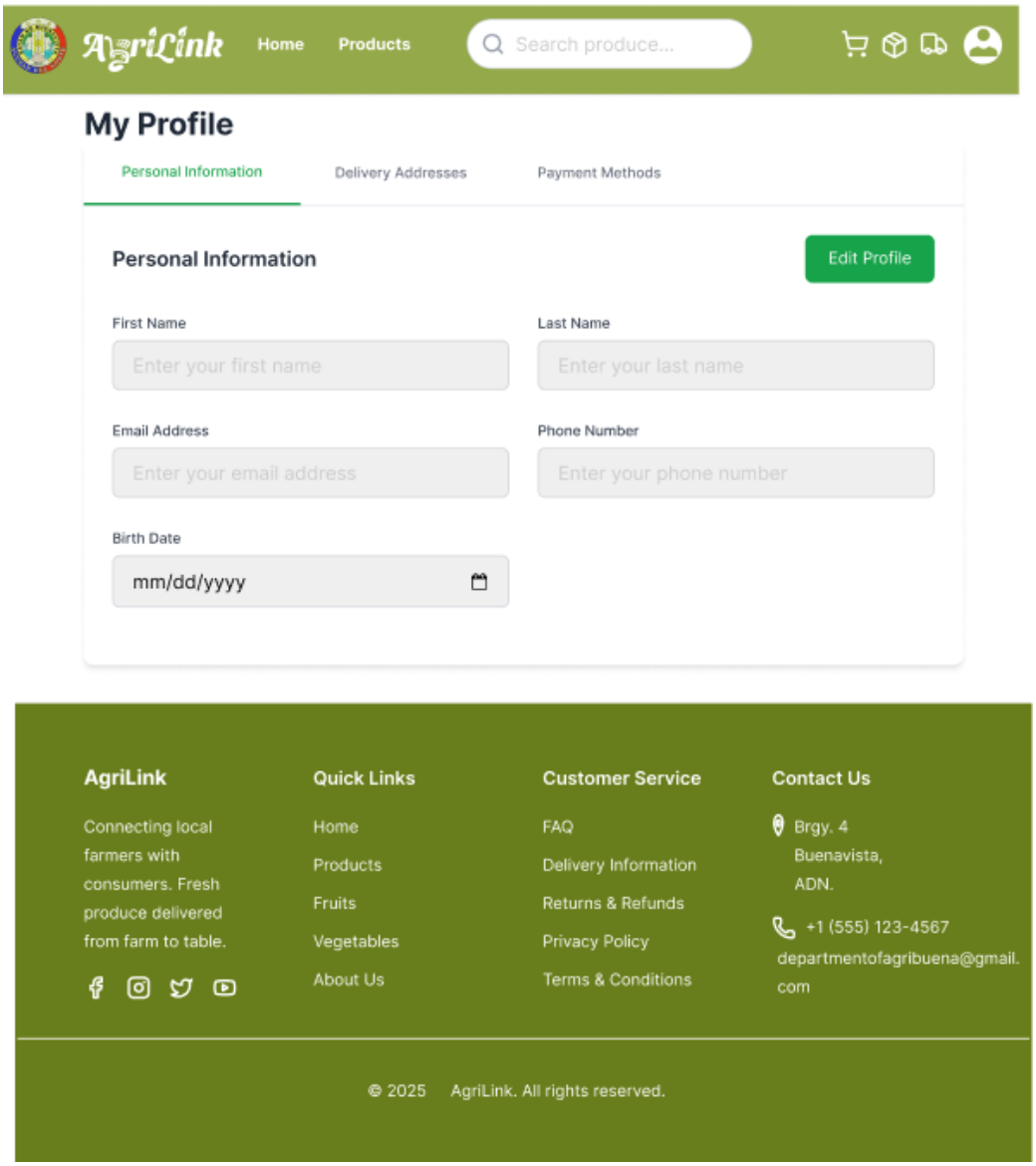


Figure 41. Buyer Profile Management

This page is the "My Profile" page, wherein consumers can edit their personal information. These sections are for personal information, including name, email address, phone number, and birth date. There are also delivery address and payment tabs that the user may set up. Through this page, every buyer gets to control their account details and settings on the Agrilink Platform.

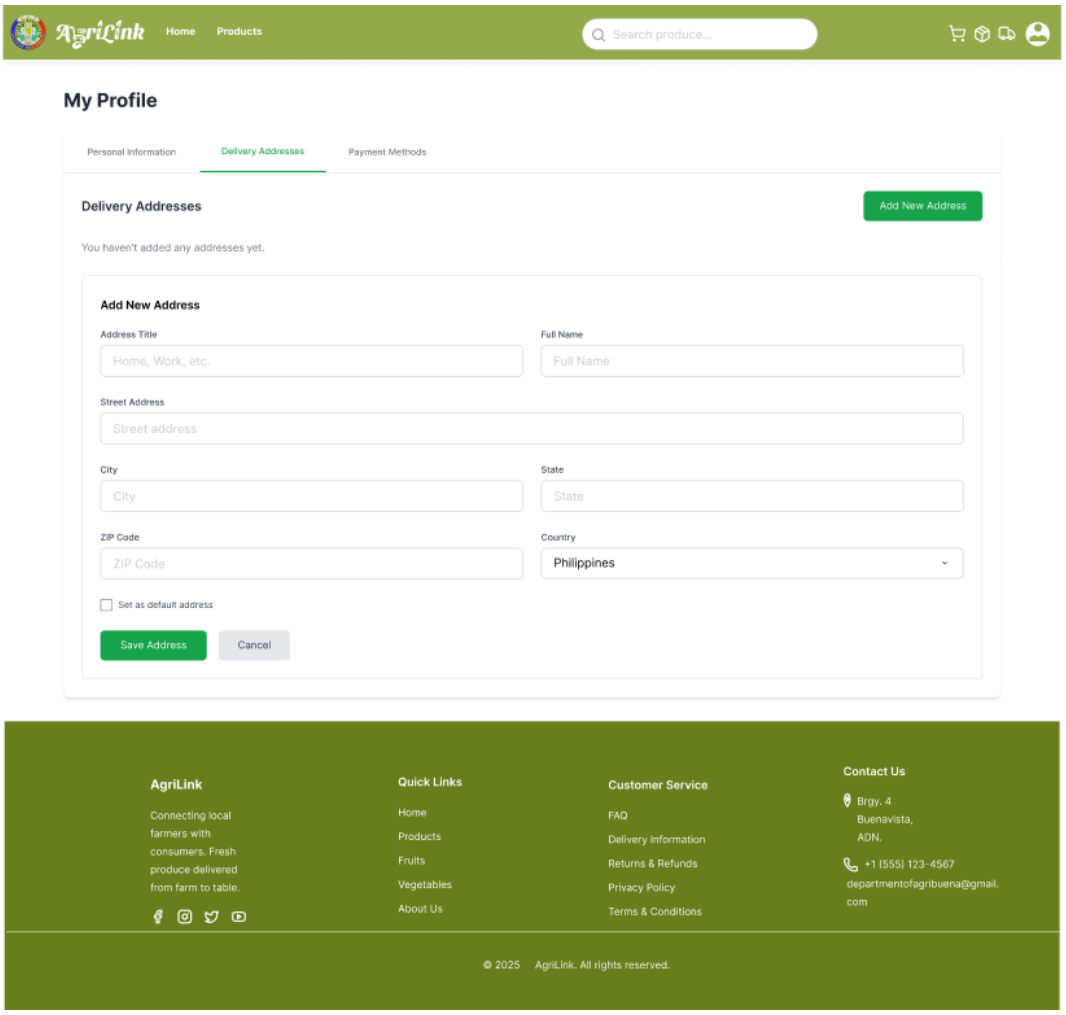


Figure 42. Delivery Address Section

This screenshot displays the Delivery Address Section of the My Profile Page, where consumers can manage their delivery addresses. It will be possible to view the option to add a new address by entering the mandatory fields, such as Address Title, Street Address, City, State, Zip Code, and Country. If no address has been saved yet, the system displays the message "You haven't added any addresses yet." With this feature, buyers can offer assurance when selecting and managing their delivery addresses.

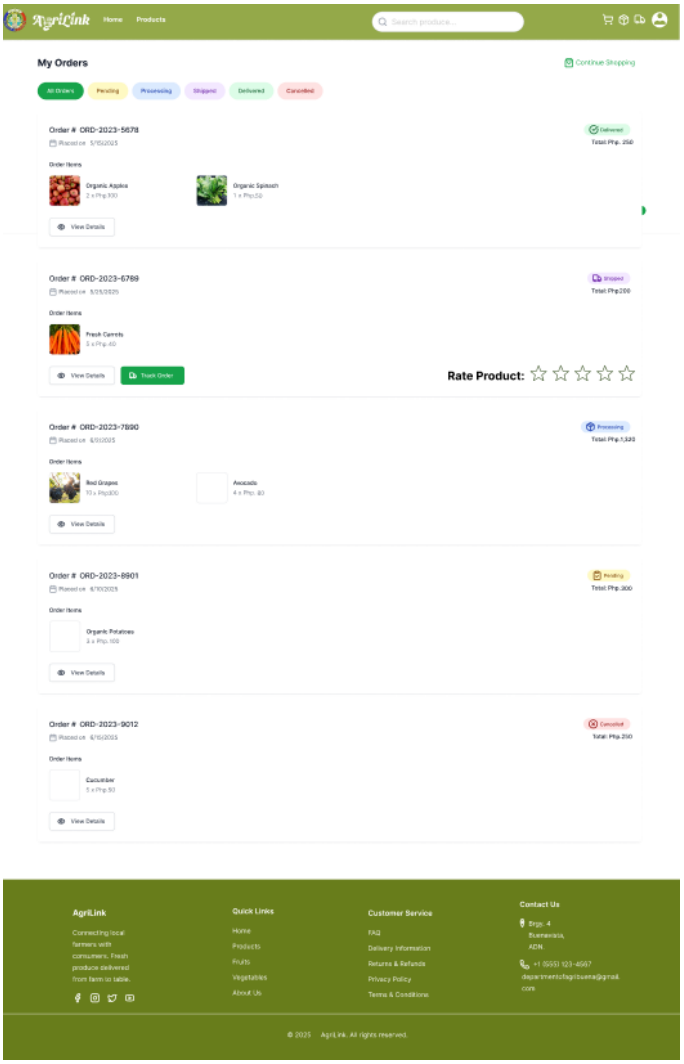


Figure 43. Order History

This is the My Orders Page, where customers can see their order history and status. There are orders with complete information, such as order number, order date, product purchased, and total amount. There are also filters, such as pending, processing, shipped, delivered, and canceled, which are used to group the views conveniently. Through this page, customers can easily track all their transactions in the Agrilink Platform.

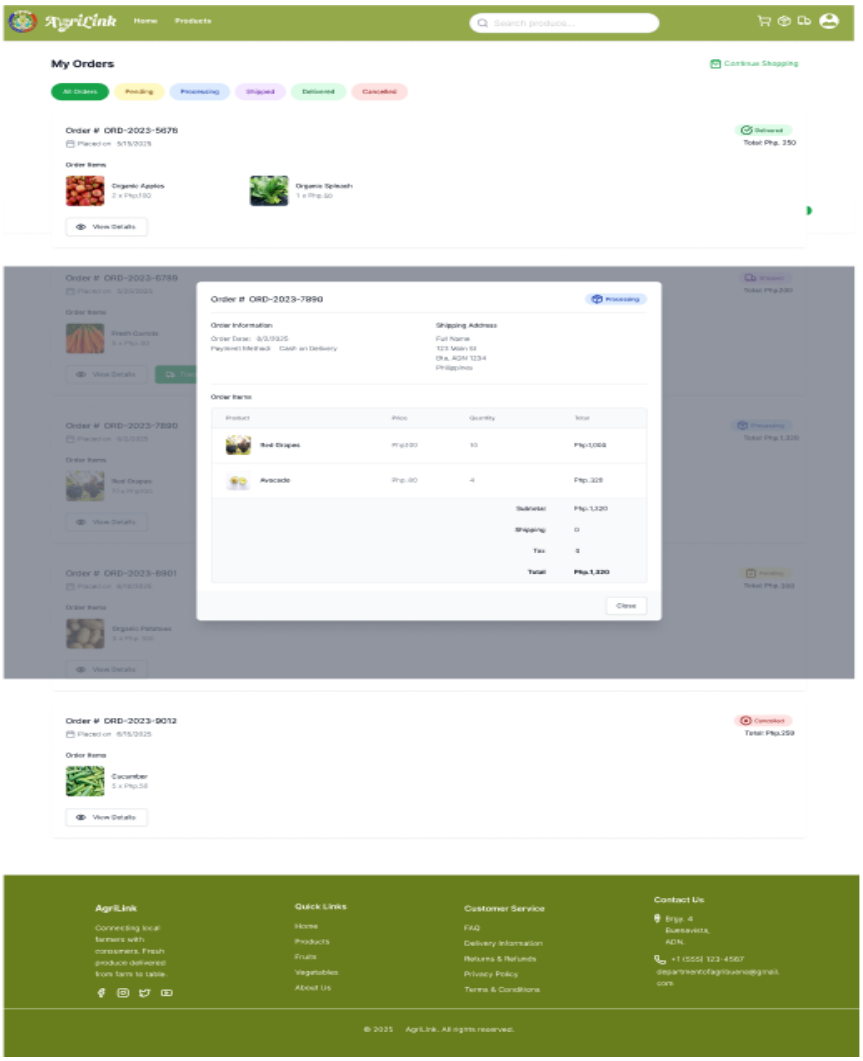


Figure 44. Transaction Details

This is the My Order Details page, wherein customers can see the complete details of their past transactions. It will retrieve order data, including order number, date, and payment method. It will include a shipping address and a comprehensive list of purchased products, along with prices, quantities, and total values. There is even an order summary that displays subtotal, shipping charge, and final amount paid. Through this page, the buyers deliver transparency and exposure of all information about their executed deals on the Agrilink Platform.

2.9 Software Platforms, Development Environments, and Tools

Table 1

Components	Specification	Usage
Internet Browser	Any (Google Chrome, Mozilla Firefox, Microsoft Edge, etc.)	A web browser allows users (farmers, buyers, and admins) to access and interact with the AgriLink E-Commerce Platform easily. It is responsible for rendering the website's interface and ensuring smooth navigation for users after they log in.
Front-end	HTML	HTML defines the structural layout of the AgriLink Platform, arranging elements such as forms, product listings, order details, and navigation menus.
	CSS	CSS is used to style and enhance the visual design of HTML structures, making the platform more user-friendly, professional, and responsive across various devices.
	JavaScript	JavaScript enhances the platform's interactivity, enabling features such as dynamic product filtering, cart updates, order tracking, and seamless user interactions.
	Bootstrap	Bootstrap provides pre-designed responsive components that allow developers to build mobile-friendly and consistent layouts for the AgriLink Platform quickly.

Back-end	**PHP**	**PHP is the primary server-side scripting language used to handle system logic, database queries, order management, authentication, and overall business processes.**
Database	**MySQL**	**MySQL is the relational database system used to store and manage farmers' products, buyer information, orders, payments, and reports efficiently.**

Table 2

2.10 Hardware Requirements

Component	Specification	Usage
Server	Intel Core i7 or higher, 16GB RAM, 1TB SSD, Gigabit LAN	The server hosts the AgriLink platform, manages the database, processes user requests, and ensures smooth operation of all system modules.
Client Computer	Intel Core i3 or higher, 4GB RAM, 500GB HDD/SSD	Farmers, buyers, and admins use client computers to access the platform via a web browser, enabling them to interact with the system effectively.
Mobile Device	Android/iOS smartphone, 3GB RAM minimum, 32GB storage, stable internet connection	Mobile devices allow farmers and buyers to access AgriLink on the go, ensuring wider accessibility and convenience for transactions.
Networking Equipment	Router, Switch, Wi-Fi Access Point	Provides reliable internet connectivity and stable network communication between clients, server, and database.
Printer/Scanner	Standard Inkjet/Laser Printer, Flatbed Scanner	Used by administrators to print transaction reports and scan important documents for record-keeping or verification.



2.11 System Evaluation

This section describes the various evaluation methods performed on the system to assess its functional suitability, performance efficiency, and usability. The survey instrument is derived from the ISO 25010 Software Product Quality Standards. The test scale provides respondents with four options and rates the system based on the question, with a score of four being the highest and one being the lowest.

Data were analyzed and interpreted based on the following parameters:

Table 3

Functional Suitability

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
4	3.50 – 4.0	Very Functional
3	2.50 – 3.49	Functional
2	1.50 – 2.49	Moderately Functional
1	1.0 – 1.49	Poor Functional

Table 3 presents the criteria used to evaluate the functional suitability of a product or system, as outlined in the ISO survey.

Table 4

Performance Efficiency

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
4	3.50 – 4.0	Very Efficient
3	2.50 – 3.49	Efficient
2	1.50 – 2.49	Moderately Efficient
1	1.0 – 1.49	Poor Efficient

Table 4 outlines the scoring parameters used to assess the system's overall performance efficiency in executing its functions and tasks.

Table 5

Usability

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
4	3.50 – 4.0	Very Usable
3	2.50 – 3.49	Usable
2	1.50 – 2.49	Moderately Usable
1	1.0 – 1.49	Poor Usable

Table 5 presents the scoring parameters used to evaluate the system's usability.

Table 6

Respondent's Distribution

AgriLink E-Commerce Platform		
Position	**N (Population)**	**Percentage (%)**
Farmer	**3**	**37.5%**
Buyer	**4**	**50%**
Admin	**1**	**12.5%**
Total	**8**	**100**

Table 6 shows a detailed breakdown of the participants who participated in the evaluation of the AgriLink E-Commerce Platform. The participants were divided according to their roles, where 37.5% were farmers and 62.5% were consumers. This indicates that most of the feedback comes from consumers, who primarily use the platform's functionalities for product search, ordering, and payment. Their evaluations provide valuable information about the usability, effectiveness, and overall efficiency of the system in facilitating direct transactions. On the other hand, farmers' opinions emphasize how the platform facilitates product management and promotes access to a wider market.

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Tertiary: Saint Michael College of Caraga

Membership/Affiliations

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Tertiary: Saint Michael College of Caraga

Membership/Affiliations

ORGANIZATION	POSITIONS HELD
N/A	N/A

Ethical Standard

A. Protection of Intellectual Property Rights (IPR)

The research shall maintain intellectual property rights by ensuring that all materials developed, including the design of the AgriLink system, interface, and source code, are copyrighted under Philippine law. All third-party software, frameworks, and data used in the system will be appropriately referenced to comply with intellectual property requirements. Two logos are also employed: the LGU Buenavista logo, officially approved for use, and the AgriLink logo, an original artwork designed by the researchers.

B. Informed Consent

Farmers, consumers, and agricultural officials who will participate in the trial of the system will be fully informed about the research objective, methods, and expected results before they participate. They will be given clear instructions about their voluntary contribution and the confidentiality of their responses. Signed consent forms are available in English and Filipino for complete understanding. Since no animal testing is involved, ethical guidance regarding consideration of animal welfare is unnecessary.

C. Data Privacy and Confidentiality

The study will comply with the Data Privacy Act of 2012 (RA 10173). Personal data collected from participants such as names, contact information, and demographic details will be kept confidential and anonymized. All electronic data will be stored in password-protected databases. Anonymized data will only be shared for academic purposes with prior consent from participants

D. Voluntary Participation

The participation in the study will be entirely voluntary. Participants will be informed that they have the right to withdraw from the research at any time without incurring any penalty or disadvantage. No undue influence or coercive measures will be used in the recruitment of participants.

E. Minimization of Harm

The research will be planned in a way that ensures that no participant will experience physical, psychological, or emotional harm. All testing will be conducted in a safe and controlled environment. Participants could stop their participation at any time if they felt uncomfortable. Since no animal testing is done, the 3Rs guideline (Replacement, Reduction, Refinement) does not apply.

F. Beneficence

The study aims to make a meaningful contribution to the agricultural industry by providing technological solutions that improve market accessibility, efficiency, and transparency between farmers and consumers. Although participants will likely not receive financial benefits, they will be part of creating a system that promotes sustainable agricultural exchange in Buenavista. A summary of the results will be presented to participants upon request.

G. Justice and Fair Participant Selection

Selection is based on unbiased and transparent inclusion criteria, ensuring representation of various groups of farmers, consumers, and agricultural officials in Buenavista. It is non-discriminatory to facilitate equal participation.

H. Data Integrity and Accuracy

Standard methods will be used to achieve accuracy and reliability in data analysis and collection. All data from system testing will be validated to avoid bias and error. All weaknesses, including sample size and external factors affecting uptake, will be acknowledged. AI tools can be used to organize and analyze the data, but researchers will analyze the final interpretations.

I. Transparency in Reporting

Standard procedures will be implemented to ensure accuracy and reliability in data collection and analysis. Validation of all data from system testing will be performed to ensure accuracy and minimize bias and error. Limitations, such as sample size or external factors influencing usability, will be considered. AI tools can be utilized to help organize and examine data, but researchers will verify the final interpretations.

J. Use of Patented or Copyrighted Materials

All findings from the research will be reported honestly, and no data will be falsified or falsely created. Proper citation will be given to all references, frameworks, and software used. In the event of any conflict of interest, it will be disclosed to ensure full transparency.

K. Ethical Considerations for Animal and Human Trials

Participants will be informed of the possible risks and benefits of testing the system before it begins. Because the research carries only minimal risk, no medical or psychological intervention is required.

L. Responsible Use of AI and Related Technologies

This capstone project received ethical approval from the Institutional Review Board of Saint Michael College of Caraga. Approval was granted on [insert date], ensuring compliance with

all necessary guidelines regarding data privacy, intellectual property, and the treatment of human participants.

M. Ethical Approval

This capstone project received ethical approval from the institutional review board of Saint Michael College of Caraga. Approval was granted on [insert date], ensuring compliance with all necessary guidelines regarding data privacy, intellectual property, and the treatment of human participants.